

When a Learned Path Integrator Makes a Map: Controlled Measurements on MapFormer

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Abstract

MapFormer builds a sense of place by accumulating a content-dependent, signed phase and applying it as a rotation in attention. We report controlled measurements of what that accumulated “where” buys when the observation map is redrawn at test, so that every evaluation measures transfer to a new instance of a known structure. On the paper’s task under a converged recipe, path integration is the larger factor on raw accuracy at every length: it adds +0.243 at training length and +0.359 at eight times it ($n = 8$, 8/8 seeds), while the rotary encoding (PoPE against RoPE) is detectably negative at training length, mostly because PoPE lowers the index code, and adds a smaller, detectable amount beyond it. Three conditions decide whether the phase pays.

- **Integrable input.** Under turn-and-forward actions the advantage over an index code falls from +0.438 to +0.050. Recording realised displacement restores it to +0.488 ($n = 8$).
- **Cancelling increments.** On the torus at eight times the training length, forcing MapWM’s increment non-negative costs 0.363 raw and 0.280 at matched loss (12/12 seeds; loss-matched in a six-arm pool, and loss-matched verdicts depend on the pool, Section 5.4). At matched loss no measurable gain over an index code remains, although raw accuracy stays detectably above it; this extends a known result to navigation, and Selective RoPE’s generator pays the same raw cost. On a counting task the raw cost is small for these two generators (0.060, unmeasured, and 0.068), but single-origin MapEM pays 0.198. The paper’s rank-2 bottleneck learns a skewed basis, and rank 4 repairs it for 384 parameters.
- **Findable solutions.** MapEM, whose shared position kernel is TEM’s conjunction of where and what, represents a varying-offset recall task exactly yet, with a single shared origin, trails MapWM by 0.375 from scratch at a shared budget ($n = 8$; 0.137 with the paper’s separate origins). On that task, constructions place the gap in search rather than representation, and training finds the solution per query token. In a separate batch, per-pair position origins add 0.215 over single-origin MapEM ($n = 48$): 0.124 from added pathway capacity with the kernel still shared, and 0.091 from per-pair freedom, both detectable.

Explicit state correction gives no measurable benefit on a drifting task built for it and helps only beyond the training length; an explicit token gate adds no measurable accuracy, and hierarchical pooling’s only detectable gain is a ceiling-level 0.012 (its larger gains, up to +0.151 on compositional transfer, are directional). Weight-shared recursion does add accuracy where the base model trains unreliably (on Match-Query, in an unpaired analysis); on the torus its gain vanishes at matched loss. Transfer across a change of structure, and language modelling, are not tested.

Contents

1	Introduction	3
2	Background: the accumulator, the kernel, and where MapFormer sits	5
3	Methods	6
4	Reproducing MapFormer	9
5	Claim 1: the accumulated phase carries transfer	10

6	Claim 2: the environment must make displacement a function of the token	14
7	Claim 3: the increment must cancel, and cancel cleanly	17
8	Claim 4: a shared kernel is representable but harder to find	21
9	Claim 5: what added machinery buys	26
10	Explicit state correction: no measurable benefit	29
11	The length axis: an open problem	30
12	Nested hierarchies: Dyck-2, and what its metric measures	31
13	PoPE’s own tasks, and why the two mechanisms anti-compose	32
14	Limitations	34
15	Conclusion	35
A	Gates and floors	36
B	Early baselines and TEM	37
C	Family tree and non-commutativity	38
D	Compositional task and hierarchy	39
E	MiniGrid and MiniWorld: supplementary	39
F	Rank and generator probes	40
G	EM kernel: supplementary	41
H	State correction: supplementary	42
I	Loops: supplementary	42
J	Language and borrowed benchmarks	42
K	The MONOTONE batch	43

1 Introduction

A model that knows *where* it is, separately from *what* it sees there, should be able to carry that knowledge into a place it has never observed. This is the claim of the Tolman–Eichenbaum Machine (Whittington et al., 2020): structural knowledge is invariant across environments, sensory content is specific to one, memory stores their conjunction, and the factorisation is what buys transfer. The research line reported here asks when a transformer’s positional mechanism can serve as that structural code.

MapFormer (Rambaud et al., 2025) carries the factorisation into attention. Each token, action or observation, contributes a learned increment to a phase; the phase is accumulated by a prefix sum and applied as a rotation. In MapFormer-WM (MapWM) the rotation acts on content-derived queries and keys. In MapFormer-EM (MapEM) it acts on two learned position vectors, producing a position kernel that multiplies a content score. The phase depends only on the token stream’s transition structure. The paper shows that both variants learn cognitive maps on grid navigation and generalise out of distribution.

Except for two appendix baselines that say so (Appendix B), every evaluation in this project redraws the observation map from a held-out seed. The binding of observations to places at test time is therefore entirely new; only structure can carry over. Every accuracy reported below is a measurement of transfer to a new instance of a known structure. None of it measures transfer across a change of structure (topology or action space), which is the stronger form of the TEM claim.

What is already known. Most of the conceptual ground is published, and the report is written against it.

- **The taxonomy.** That a rotary logit has two slots, a phase and a magnitude, and that translation-invariant interval operators form a one-parameter group whose Jordan form yields decay, rotation and polynomial terms, is due to GRAPE (Zhang et al., 2026) and Puranik (2026), who gives the Jordan-form reason it is exhaustive; Vetcha (2026), posted between the two, states the two-slot decomposition independently.
- **Content-dependent rotation.** Mamba-3 (Lahoti et al., 2026) publishes a data-dependent complex state whose equivalence to data-dependent RoPE is its Proposition 3. Selective RoPE (Movahedi et al., 2026) places a content-dependent cumulative sum in the phase and was posted three days before MapFormer; neither paper cites the other. A long-context survey (Liu et al., 2025) already has a “content-aware position embedding” category, whose members are CoPE (Golovneva et al., 2024) and DAPE (Zheng et al., 2024).
- **The sign of the increment.** That non-negative state transitions cannot track parity and that negative values restore it is due to Sarrof et al. (2024) and Grazzi et al. (2025); Selective RoPE’s Section 4.2 carries it into rotations in attention.
- **Contextual counting.** That a fixed index code cannot count contextually and a content-gated position can is CoPE’s claim (Golovneva et al., 2024).
- **EM and WM.** MapFormer’s two variants follow Whittington et al. (2025), which argues the two algorithms reach equivalent solutions with different learning dynamics, and TEM-t (Whittington et al., 2022) relates TEM to transformers.
- **Recursion.** That a weight-shared block applied repeatedly can substitute for depth is the premise of the recursive-transformer line, including Mixture-of-Recursions (Bae et al., 2025).
- **Rank.** The nearest published neighbour of the rank question is LieRE (Ostmeier et al., 2025), a learned skew-symmetric generator basis driven by known position.
- **Language.** Content-dependent phase has published negatives on language modelling (HGRN, Qin et al., 2023, Table 11; HGRN2, Qin et al., 2024, Table 1), and MapFormer v4’s own language experiment reports a perplexity gain but no BLiMP gain.

The argument of this report. On tasks where the observation map is redrawn at test, the transferable part of MapFormer is its accumulated, content-driven position phase. On MapFormer’s own task it is the larger factor on raw accuracy at training length under the paper’s recipe and at every length under

Table 1: Contributions and their standing, one result per row. *Standing:* “ours” = no paper in the local corpus (`papers/INDEX.md`) reports it; “replication” = a published claim repeated here. *Prior work:* closest published claim. Path integration = MapFormer’s position, a running sum of learned per-token phase increments; index code = position given by token index; monotone = increments forced non-negative; r = rank of the token-to-increment bottleneck; loss-matched = compared at equal final training loss. Reading: the sign result (in navigation) and contextual counting replicate published claims; no corpus paper reports the others.

result	standing	prior work
Path integration is the larger factor on MapFormer’s own task on raw accuracy: at training length under the paper’s recipe, and at every length measured under a converged recipe (index and PoPE controls, context destruction); the encoding adds a smaller amount beyond training length	ours	MapFormer shows path integration solves its task
Rotation-based actions remove the effect; recording realised displacement restores it; the MiniGrid flip needs $r = 4$; the aliasing account fails on its converged cells (exploratory at the least aliased level)	ours	none evaluates navigation
Monotone increments destroy the map in navigation (at matched loss in a six-arm pool, no gain over an index code remains; loss-matched verdicts depend on the pool), isolated as one operation; a second generator pays the same raw cost	replication in a new regime	Sarrof; Grazzi; Selective RoPE §4.2
The rank-2 bottleneck learns a skewed basis; $r = 4$ repairs it	ours	LieRE (different axis)
Index codes cannot count contextually	replication	CoPE
An unconstrained MapFormer counts through a content gate, by intervention	ours	CoPE’s gate is the analogue
Shared versus per-pair position kernel: existence, holdability, per-token search, per-pair decomposition	ours	Whittington et al. (2025) and TEM are the frame
State correction shows no detectable scaling with drift; an explicit gate separates and buys no measurable accuracy; hierarchy directional	ours	–
Recursion adds accuracy to path integration on one task (unpaired), with a super-additive paired estimate; recursion’s torus gain is convergence	ours	recursion for depth: Mixture-of-Recursions

a converged one; the rotary encoding also matters beyond the training length (Section 5.1). That phase pays under three conditions, each supported by an intervention on its own task; that rank acts through conditioning is descriptive. The input must make a token’s displacement fixed; the increment must be signed and well-conditioned so that it can cancel; and training must find the solution. Fully factorising where from what, as MapEM’s shared kernel does, costs nothing measurable in what can be represented on the tasks tested. On a varying-offset recall task it detectably changes what training finds, and per-pair position freedom detectably raises accuracy on it. Explicit correction and gating add no measurable accuracy on these tasks at training length, and pooling’s only detectable gain is ceiling-level (its larger gains are directional). All of this is transfer to a new instance of a known structure.

The three conditions share one mechanism statement. Attention reads the interval sum of per-token increments. The first condition asks whether a fixed per-token increment can equal the displacement; the second, whether the interval sum cancels so that it measures net displacement; the third, whether gradient descent reaches increments that do so. The conditions are not claimed to be sufficient, and they are not an exhaustive list. They were established on different tasks, and no single task shows all three binding at once.

Table 1 lists what is new here and what replicates prior work.

Organisation. Section 2 sets out MapFormer’s mechanism and the position kernel. Section 3 gives tasks, validity gates, statistics and recipes. Section 4 reports the reproduction. Sections 5–9 give the five claims, Section 10 the state-correction result, and Section 11 the open problem of out-of-distribution length. Sections 14–15 give limitations and conclusions. Supplementary measurements are in the appendices.

2 Background: the accumulator, the kernel, and where MapFormer sits

2.1 The accumulated phase

MapFormer maps every token x_t (action or observation) through a rank- r bottleneck to a phase increment and accumulates it:

$$\Delta_t = W_{\text{out}} W_{\text{in}} x_t \in \mathbb{R}^{n_b}, \quad S_t = \sum_{u \leq t} \Delta_u, \quad \theta_t = \omega \odot S_t, \quad (1)$$

with $W_{\text{in}} \in \mathbb{R}^{r \times d}$, $W_{\text{out}} \in \mathbb{R}^{n_b \times r}$, n_b frequency blocks and learned angular velocities ω . The paper uses $r = 2$, justified by the two-dimensional displacement vector. Index RoPE (Su et al., 2021) is the special case $\Delta_t \equiv 1$. Because the prefix sum is linear, every phase channel is a linear readout of one r -dimensional accumulator, and the attention between positions t and s depends on $S_t - S_s$, the interval sum of the increments between them.

Frequency initialisation. The paper initialises ω geometrically between ω_{max} and $\omega_{\text{max}}/\Delta_{\text{max}}$. Its printed formula (v4, eq. 18) has an exponent $-i/n_b$ that makes ω_i increase with i , contradicting its own boundary condition. All runs here use $\omega_i = \omega_{\text{max}}(1/\Delta_{\text{max}})^{i/(n_b-1)}$, which decreases monotonically.

2.2 Two ways the kernel meets content

MapEM. Attention is $\text{softmax}(A_X \odot A_P)$, a Hadamard product of a content score A_X and a position score built from two learned vectors,

$$A_P[t, s] = q_0^\top R(\theta_s - \theta_t) k_0 = \sum_i a_i \cos(\omega_i(S_t - S_s) + \phi_i) =: \kappa(\Delta S), \quad (2)$$

with $a_i = |q_{0,i}| |k_{0,i}|$ and per-block phase ϕ_i . One kernel κ serves every query-key pair; content cannot reshape it. The paper’s design uses separate initial vectors q_0 and k_0 (App. A.7 of v4, where the authors “suspect this separation to be beneficial”). The ablation used throughout is a single shared origin p_0 ($q_0 = k_0 = p_0$, so $\phi_i = 0$). The Hadamard product is exactly TEM’s conjunction: $(A_X \odot A_P)_{ts} = (q_t^x \otimes q_t^p) \cdot (k_s^x \otimes k_s^p)$, one bilinear form on the tensor product of content and position. The sign of κ is a gauge: $A_X \odot (-\kappa) = (-A_X) \odot \kappa$, so a negative content gain turns retrieval at the kernel’s peak into retrieval at its trough.

MapWM. The rotation acts on content-derived Q, K : $Q_t^\top R(\theta_s - \theta_t) K_s = \sum_b |Q_b| |K_b| \cos(\omega_b \Delta S + \phi_b(q, k))$. Amplitudes and phases are set per block *and per pair* by content. MapWM is therefore not an additive combination of a content and a position score; its position kernel is per-pair.

Three axes. Equations (1)–(2) separate three design choices: the *argument* (what ΔS measures: signed or monotone increments, the rank r , or the index); the *kernel* (the shape of κ : phases, magnitudes, frequencies); and the *composition* (one shared κ multiplying content, or a per-pair kernel). Claims 2 and 3 are about the argument, Claim 4 about composition.

2.3 Where the mechanisms sit

Table 2 places MapFormer among the mechanisms it is most often compared to. The entries are from the local corpus (papers/INDEX.md).

MapFormer v4. The reproduction target changed between versions. Version 1 of Table 2 (2D columns) lists MapWM 0.99/0.99/0.96 and MapEM-os 1.0/0.99/0.97 for IID, OOD-d and OOD-s; version 4 lists 1.00/1.00/0.99 for MapWM-r2 and 1.00/1.00/1.00 for MapEM-os. Version 4 also reports that MapWM collapses on a 5D grid (0.75/0.50/0.35, Table 6) and adds a language experiment: RoPE 19.14 ± 0.14 against MapWM 18.79 ± 0.15 perplexity on OpenWebText, with no gain on BLiMP.

Table 2: Positional mechanisms by slot, driver and sign of the increment. *Slot*: which part of the rotary logit changes, phase (rotation angle) or magnitude. *Sign*: sign of the per-token increment Δ accumulated into position; signed increments can cancel, +1 is a clock, “-” = no accumulated per-token phase. Reading: MapFormer, Selective RoPE and Mamba-3 accumulate a signed phase; RoPE, CARoPE and CoPE only count upward.

mechanism	slot	what drives it	sign
RoPE	phase	the index ($\Delta \equiv 1$)	+1, a clock
MapFormer (WM, EM)	phase	content, rank- r bottleneck, prefix sum	signed
Selective RoPE	phase	query, causal convolution, sigmoid gate, no bottleneck	signed
CARoPE (Veisi et al., 2025)	phase	content, one scalar per head	$1/(\text{softplus}+1) \in (0, 1)$
Mamba-3	phase and magnitude	data-dependent complex state	signed
PoPE (Gopalakrishnan et al., 2025)	magnitude	content sets $\mu = \text{softplus} \geq 0$; phase is the index	-
MapPoPE (this project)	both	PoPE magnitude plus MapFormer phase	signed
FoX (Lin et al., 2025)	magnitude	per-head sigmoid forget gate, log-cumsum bias	-
CoPE	outside the per-token frame	query-key sigmoid gate; no per-token θ_t exists	monotone
PaTH (Yang et al., 2025)	outside the frame	cumulative product of Householder transforms	-

3 Methods

3.1 Models and knobs

All models are small: one to four layers, two heads, $d = 128$ on the torus tasks (about 204k parameters for one layer). Arms compared in one table are parameter-matched unless a count is given. The main arms, named as in the code:

- **Vanilla** (MapWM, $r = 2$) and **Vanilla_r4 ... Vanilla_r32** (only r differs); **VanillaEM** (MapEM, separate q_0, k_0); **VanillaEM_P0** (single p_0); an **_r4** suffix sets $r = 4$.
- **Index baselines**: **RoPE** (MapFormer’s architecture with index rotation), **PlainFlat** (an ordinary index-RoPE transformer), **PoPE-Flat** (index phase with PoPE magnitude), and **MapPoPE-Flat** (PoPE magnitude with MapFormer’s phase).
- **Sign knob**: **Signed_r4** (Δ as in eq. 1), **Abs_r4** ($|\Delta|$), **Pos_r4** (softplus, GRAPE-style), **CARoPE_r4** (CARoPE’s $1/(\text{softplus}+1)$), each with 204,757 parameters.
- **Recursion**: **Looped** applies one shared block four times, at the parameter count of one layer; **LoopedSampled** draws the count per batch.
- **State correction**: **Level15**, a wrapped invariant Kalman filter on the path integral with a constant prior variance and a per-token measurement noise (Section 10).
- **Gates**: **Gated_r4** multiplies Δ by a per-token sigmoid (CoPE’s selection on MapFormer’s direction); **Forget** adds a FoX-style learned decay bias.
- **EM kernel variants**: **EMPair_r4** (per-pair origins), **EMPairConst_r4** (its capacity control), **EMDoF_alignfree** and **EMDoF_magonly** (phase freedom and its matched control), installed-rewind arms (Section 8.4).

The index baseline’s frequency schedule. The repository’s RoPE originally used $\text{base}^{-c/(n_b-1)}$ rather than the canonical schedule. On parity at $n = 16$, canonical minus repository is inside its MDE at every

Table 3: Main-text tasks. *Chance / floor*: chance = one over the number of answers; “floor” is a measured no-model baseline on the scored events; multiple floors are for MiniGrid at $T = 128/512/1024$ and MiniWorld at $16/64/256$ observation types. *Train / eval length*: sequence length in training / at evaluation. *Gates passed*: n -grams on the action or answer stream alone sit at chance or floor; context destruction collapses trained models. Gate details are in Appendix A. The compositional motif task and the family tree appear in Appendices C and D only. Reading: every task evaluates on held-out episodes and, where there is an observation map, on a map drawn from a held-out seed.

task	tests	chance / floor	train / eval length	gates passed
Paper task (torus 64×64 , 16 observation types, $p_{\text{empty}} = 0.5$, loss at revisits)	in-context map, revisit prediction	blank floor 0.506 (0.522 on the OOD evaluator’s events)	$T = 128 / 128$, 512, 1024	context destruction (Sec. 5.1)
Knob torus (turn/forward actions, 4 or 12 headings)	whether a fixed per-token increment can name a displacement	floor 0.508	$T = 128 / 128$	answer-stream n -grams before training
Match-Query 64^2 , 128^2	blind continuation: observations withheld in the query phase	chance 0.0625; never-moved floor 0.0893 (64^2)	explore 512, query 256	n -grams; context destruction
Parity	a signed prefix sum as a register	chance 0.5	$L = 16 / 16$ –256	trivial baselines within +0.0122
Recency (k -back with uncounted filler)	contextual position; varying retrieval offset	chance 0.0625; most-recent floor 0.0771	$T = 1024 / 1024$, 2048	n -grams; oracle; token distance
MiniGrid DoorKey-16x16 (egocentric)	an external benchmark, commanded or allocentric actions	floors 0.635 / 0.536 / 0.490	$T = 128 / 128$, 512, 1024	floors
MiniWorld OneRoom (location-keyed tokens, fresh map)	map extent and aliasing	non-blank marginal 0.077 / 0.031 / 0.013	$T = 512 / 512$	n -grams; label mass; lag; context destruction

length, the tightest being -0.0023 (MDE 0.0086) at $L = 256$; the code was switched to the canonical schedule. RoPE checkpoints trained before the switch keep their own schedule.

3.2 Tasks

Table 3 lists the tasks used in the main text. Every task evaluates on held-out episodes and, where there is an observation map, on a map drawn from a held-out seed.

Paper task. MapFormer’s 2D grid navigation: a random walk on a 64×64 torus emits an interleaved stream of actions and observations, 16 observation types with half the cells blank, and the loss is taken only at observations of revisited cells. The paper’s OOD protocol (App. B) varies grid size, blank probability and length (Section 4).

Match-Query. The model explores with observations revealed, then continues blind: the query phase has actions but no observations, and the model must predict the observation at each revisited cell. It closes the retrace route that lets an index model exceed the floor on the paper task (Section 5.1).

Recency. The model must retrieve the k -th most recent content symbol ($k \leq 64$, 16 symbols) while filler tokens are interleaved and not counted ($p_{\text{filler}} = 0.5$). At $k = 64$ the answer sits 128.3 ± 11.8 tokens back, so k is a contextual rather than an index offset. Recency has no observation map; held-out episodes are new symbol sequences.

3.3 Validity gates

A task is used only after it passes gates run before training on the task code itself.

- **Shortcut gates.** n -gram predictors of orders 1–5 on the action or answer stream alone, the marginal, and task-specific trivial predictors (never-moved, most-recent, last-bit) must sit at the chance or floor rate. Two task families failed these gates and are excluded (Appendix A).
- **Context destruction.** On trained models, the context the task claims to require is destroyed and accuracy must collapse. Two forms are reported: a shuffle, and an on-manifold resample from an independent episode. They order differently by task (resample is more destructive on the paper task and less destructive on the compositional task), so both are reported.
- **Measured floors.** Every table states the floor measured on its scored events, not only 1/vocabulary.
- **Manipulation checks.** Arms that differ in one operation are verified bitwise identical at initialisation or with the operation disabled, constraints are verified to hold after training, and gradient masks are verified at every epoch.

3.4 Statistics

- **Paired contrasts.** Contrasts are differences paired by seed. The minimum detectable effect is $\text{MDE} = 2.8\text{sd}/\sqrt{n}$ on the paired differences. A contrast outside its MDE is *detectable*; inside it, *unmeasured*, never “null”. A negative result is called *powered* only when its MDE is smaller than the effect it is used to dismiss. Seed counts with a positive difference are given as, e.g., 8/8.

Worked example (the position main effect of Table 9 at $r = 2$, $T = 128$). For each seed, the effect is $\frac{1}{2}[(\text{Vanilla} - \text{RoPE}) + (\text{MapPoPE-Flat} - \text{PoPE-Flat})]$, with every arm scored on the same evaluation trajectories. Over the eight seeds this gives 0.293, 0.218, 0.240, 0.258, 0.245, 0.223, 0.288 and 0.177: mean 0.243, standard deviation 0.038. The standard error of that mean is $0.038/\sqrt{8}$, and the MDE multiplies it by 2.8, the factor that flags a true effect of that size at least 80% of the time while flagging pure noise only 5% of the time ($1.96 + 0.84$). Here $\text{MDE} = 2.8 \times 0.038/\sqrt{8} = 0.038$; with eight seeds $2.8/\sqrt{8} \approx 0.99$, so the MDE is close to the sd of the paired differences. The effect is about six times its MDE, so it is detectable. Because arms are paired by seed and share evaluation data, noise common to both arms cancels in the difference, which keeps the sd, and so the MDE, small.

- **Seeds.** Results at $n \leq 3$ are exploratory whatever their verdict column says.
- **One batch.** Arms that are compared were trained in one batch with the same code. A same-seed retrain that reproduces a number bitwise is a determinism check, not a replication.
- **Fresh-seed replication.** Where a batch was extended, the new seeds are reported alone beside the pooled estimate. The first eight seeds overestimated an effect at least four times in this line, including the EM origin contrast (+0.237 at $n = 8$, +0.073 on fresh seeds), phase freedom (+0.213 on seeds 0–7, +0.113 fresh) and the per-pair total (+0.280 at $n = 8$, +0.215 pooled at $n = 48$).
- **Accuracy and training loss (“rule 9”).** For each batch the correlation $r(\text{final loss}, \text{accuracy})$ is reported. Where it is strong, contrasts are given both raw and *loss-matched*: the residual after a pooled regression of accuracy on final training loss. Loss-matching is a regression control, not randomisation, and it conditions on a mediator: when a manipulation causes the loss gap, loss-matching removes part of the manipulation’s own effect. A pool in which loss does not vary independently of the manipulation cannot identify the fit at all (Section 7.1).
- **Pre-registration.** Most batches from September 2026 were pre-registered. Verdicts are reported as registered, including when a convergence gate failed or a discriminator was mis-designed, and post-hoc analyses are labelled.
- **Readouts and symmetries.** Mechanism readouts are built to be invariant to the model’s symmetries: shifts of a block’s phase by $2\pi/\omega_i$, and the sign gauge of MapEM’s kernel. A readout that is not invariant can report the absence of a solution that is present (Section 8.5).

Table 4: Training recipes. “Paper” is the published MapFormer recipe. *Recipe*: epochs $\times$ batches per epoch $\times$ batch size, schedule and peak learning rate (lr). *Reading*: the recipe matters on these tasks as much as any architectural ingredient.

name	recipe	used in
paper	16 epochs $\times$ 98 batches $\times$ 128, linear decay from step one, lr 3×10^{-4}	Tab. 8, knob sweep
paper, 50 ep torus	50 epochs, 5% warmup + cosine to 10% 300 epochs $\times$ 98 batches $\times$ 128, 5% warmup + cosine, lr 10^{-3}	reproduction rerun sign, rank, gates, converged paper-task factorial, EM kernel on torus
recency	300 epochs $\times$ 48 batches $\times$ 16, cosine, lr 10^{-3} , 1 layer	Claims 1, 3, 4
Match-Query	300 epochs, 48 batches $\times$ 16, warmup + cosine, lr 3×10^{-4}	Claims 1, 5
MiniGrid	50 epochs $\times$ 156 batches, 3 layers, 25k cached trajectories	Claim 2
MiniWorld	400 epochs (800 for one endpoint), 5% warmup + cosine	Claim 2

3.5 Recipes

Table 4 gives the training recipes. The recipe matters on these tasks as much as any architectural ingredient: on the compositional task, changing from the published recipe to cosine decay at a higher learning rate and three times the epochs adds +0.160 (MDE 0.140, 7/8), more than any architectural effect measured there (Appendix D). On the torus, raising the learning rate from 3×10^{-4} to 10^{-3} at 300 epochs cuts **Vanilla**’s seed sd from 0.096 to 0.028 at $T = 512$ ($n = 8$), but on Match-Query the same rise, with the epochs doubled to 600, does not reduce variance (sd 0.263 to 0.261), and the compositional recipe change above roughly doubles it (sd 0.070 to 0.126).

Match-Query reproducibility. Match-Query does not reproduce across batches: the same path-integrated one-layer arm scored 0.456 in one batch and 0.416 in another under the same recipe. Retraining the looped arm with the same seeds moved individual seeds by 0.185 on average (seed 2: 0.772 to 0.123), so a seed does not determine the run and seed-paired contrasts on Match-Query are unreliable; where an unpaired analysis exists it is cited. The torus retrains bitwise across batches.

4 Reproducing MapFormer

The reimplementaion is consistent with the paper’s version 1 within about one standard deviation, sits below version 4, and confirms the parallel-scan cost claim. Two of the paper’s conjectures do not survive on map tasks.

Table 2. Table 5 compares our measurements with both versions of the paper. Our EM is MapEM-os with a single shared origin p_0 (**VanillaEM.P0**), the ablation of the paper’s separate q_0, k_0 . WM’s shortfall against the paper does not move with budget: IID 0.969 at 16 epochs and 0.968 at 50 epochs. The floors at $p_{\text{empty}} = 0.8$ are about 0.80, so the OOD-s columns have little headroom.

The navigation interpretability figure. Version 4’s Figure 9 (App. C.1; Figure 4 in version 1) makes four checkable claims about the learned code. Table 6 reproduces them on eight seeds per arm. C1 reproduces. C2 holds in direction at $r = 2$ with high variance. C3, the paper’s own reported limitation (actions that should be orthogonal are not), reproduces at $r = 2$ and is removed at $r = 4$ with no regulariser, where the paper proposes a bounded-energy constraint. C4 does not reproduce on either backbone: the ratio is below 1 in every cell. The version 4 App. C.3 passage on EM’s “two separate pools of neurons” motivated testing EM for C4; one of eight readings of C4 crosses 1 and none is much larger.

Parallel scan. Table 7 measures the paper’s cost claim on wall clock. Forward plus backward time grows 2.5 $\times$ for **Vanilla** from $L = 128$ to $L = 2048$, against 14.4 $\times$ for the non-commutative MapEM-NC-L and

Table 5: Reproduction of MapFormer Table 2 (2D columns): accuracy predicting the observation at revisited cells of a held-out map. IID = training distribution; OOD-d = smaller map, denser observations; OOD-s = larger map, sparser observations ($p_{\text{empty}} = 0.8$), at episode length $l = 512$ as in the paper’s Appendix B. MapWM = MapFormer-WM; MapEM-os (p_0) = MapFormer-EM with one shared origin p_0 instead of the paper’s separate q_0, k_0 . Paper v1 and v4 values as printed; ours are mean $\pm$ sd over 8 seeds per arm, the 16-epoch and 50-epoch budgets each one batch. Measured floors (always-blank): IID 0.522, OOD-d 0.216, OOD-s ($l = 512$) 0.799. The v4 Table 2 caption gives OOD-s at $l = 256$, where ours are MapWM 0.978 ± 0.018 / 0.984 ± 0.020 and MapEM-os (p_0) 0.988 ± 0.007 / 0.988 ± 0.012 (16 / 50 epochs; floor 0.803). Reading: consistent with version 1 within about one standard deviation, below version 4; WM’s shortfall does not move with budget.

model	condition	paper v1	paper v4	ours, 16 ep ($n = 8$)	ours, 50 ep ($n = 8$)
MapWM	IID	0.99	1.00	0.969 ± 0.037	0.968 ± 0.051
MapWM	OOD-d	0.99	1.00	0.958 ± 0.048	0.943 ± 0.076
MapWM	OOD-s, $l = 512$	0.96	0.99	0.943 ± 0.033	0.964 ± 0.029
MapEM-os (p_0)	IID	1.0	1.00	0.987 ± 0.009	0.985 ± 0.023
MapEM-os (p_0)	OOD-d	0.99	1.00	0.983 ± 0.010	0.980 ± 0.031
MapEM-os (p_0)	OOD-s, $l = 512$	0.97	1.00	0.978 ± 0.011	0.978 ± 0.015

Table 6: The four checkable claims of the navigation interpretability figure (v4 Fig. 9, v1 Fig. 4), read from trained torus models; mean $\pm$ sd over 8 seeds per arm. $D_{\text{act}}, D_{\text{obs}}$ = learned phase increments of action and observation tokens; left, right, up = increments of those actions; $v_{\text{obs}}, v_{\text{act}}$ = attention value vectors. C1: actions move position more than observations; C2: opposite actions cancel; C3: orthogonal actions are not orthogonal (the paper’s own limitation); C4: observations write content more than actions. MapEM has separate origins; r = bottleneck rank. Reading: C1 reproduces; C2 holds in direction at $r = 2$ with high variance; C3 reproduces at $r = 2$ and is removed at $r = 4$; C4 does not reproduce (below 1 in every cell).

claim	quantity	paper	MapWM $r = 2$	MapWM $r = 4$	MapEM $r = 2$	MapEM $r = 4$
C1	$\ D_{\text{act}}\ /\ D_{\text{obs}}\ $	$\gg 1$	25.0 ± 22.4	35.9 ± 19.0	21.8 ± 25.2	26.1 ± 13.0
C2	$\cos(\text{left}, \text{right})$	-1	-0.729 ± 0.373	-0.996 ± 0.004	-0.352 ± 0.896	-0.996 ± 0.002
C3	$ \cos(\text{left}, \text{up}) $	$\gg 0$ (limitation)	0.779 ± 0.276	0.174 ± 0.083	0.868 ± 0.214	0.190 ± 0.115
C4	$\ v_{\text{obs}}\ /\ v_{\text{act}}\ $	$\gg 1$	0.57 ± 0.06	0.60 ± 0.04	0.90 ± 0.21	0.64 ± 0.04

120.4 $\times$ for a faithful TEM. The models are not parameter-matched and only the growth is compared; the parallel models are overhead-dominated below $L \approx 1024$, and TEM’s constant includes Python-loop overhead.

Two conjectures. (i) Separate q_0, k_0 is not a general improvement. On map tasks it is worse: paper task 0.901 ± 0.102 (fresh map) against single p_0 ’s 0.987 ± 0.012 ($n = 3$); on Match-Query single minus separate is $+0.358$ (3/3, $n = 3$); and on the torus recipe separate minus single -0.154 (MDE 0.130, 0/8) at $T = 1024$. On varying-offset recency it is directionally better (Section 8.2). (ii) Non-commutativity is real but small where the paper motivates it. On a depth-5 family tree whose measured non-commutativity is 1.000, the non-commutative MapEM-NC-NL beats its commutative control by $+0.013$ (MDE 0.008, 3/3), while path integration beats an index code by $+0.205$ (MDE 0.118, 3/3); both at $n = 3$, exploratory (Appendix C).

5 Claim 1: the accumulated phase carries transfer

On four tasks with a redrawn map or held-out episodes, an index code falls well short of a path-integrated code, and on the paper task it sits near its floor under the paper’s recipe; on MapFormer’s own task path integration is the larger factor on raw accuracy at every length. Under a converged recipe the position main effect is $+0.243$ (MDE 0.038, 8/8) at training length and $+0.359$ (MDE 0.106, 8/8) at eight times it; the encoding main effect is -0.049 and $+0.189$ at the same lengths, both detectable. Under the paper’s recipe, with the index arms near the floor, the two effects read $+0.461$ and $+0.003$. On blind continuation, a path-integrated model scores 0.456 against an index model’s 0.108 ($n = 8$ each; paired estimate $+0.348$, but pairing is unreliable on that task); on parity it helps at every length on 8/8 seeds; on recency it adds $+0.750$ (MDE 0.030, 8/8). Raw, the torus separation holds at every length under a converged recipe; whether it survives loss-matching depends on the pool used (Section 5.4).

Table 7: Forward plus backward time (ms), batch 4, $d = 128$, 2 layers, median of 15 repetitions, at sequence length L ; growth = time at $L = 2048$ over time at $L = 128$. **Vanilla** = MapFormer-WM (parallel prefix sum); **VanillaEM_P0** = MapFormer-EM, one origin; **PlainFlat** = ordinary index-RoPE transformer; MapEM-NC-L = non-commutative MapEM (sequential product of generators); TEMFaithful = a faithful Tolman–Eichenbaum Machine. Not parameter-matched: only growth is compared; parallel models are overhead-dominated below $L \approx 1024$. Reading: the parallel models grow 2.5–3.9 $\times$, against 14.4 $\times$ for MapEM-NC-L and 120.4 $\times$ for TEM.

model	parameters	$L = 128$	$L = 2048$	growth
Vanilla (MapWM)	405,472	4.8	12.1	2.5 $\times$
VanillaEM_P0	405,600	4.3	16.7	3.9 $\times$
PlainFlat (index)	405,024	4.2	11.4	2.7 $\times$
MapEM-NC-L	430,016	28.5	410.1	14.4 $\times$
TEMFaithful	20,705	163.2	19646.3	120.4 $\times$

Table 8: Paper task under the paper’s recipe (16 epochs, linear decay): accuracy predicting the observation at revisited cells of a held-out map, at the training length $T = 128$; mean $\pm$ sd, $n = 8$ per arm, one batch; blank floor 0.506. *Encoding:* RoPE = rotation only; PoPE = content-dependent magnitude plus rotation. *Position:* index = token index; path-integrated = running sum of learned per-token increments (MapFormer). **Vanilla** = MapFormer-WM; **VanillaEM_P0** = MapFormer-EM, one origin; RoPE = MapFormer with index rotation; **PlainFlat** = ordinary index-RoPE transformer; MapPoPE is at $r = 2$. Main effects are paired by seed over MapPoPE-Flat, **Vanilla**, RoPE, PoPE-Flat: position = path-integrated minus index, averaged over encodings; encoding = PoPE minus RoPE, averaged over position. MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$; inside it = unmeasured, not evidence of zero; 5/8 = seeds with a positive difference. No sd or MDE was recorded for the position effect (per-arm sds at most 0.043). Index arms are undertrained under this recipe; the converged rerun is Table 9. Reading: index models sit just above the floor and path-integrated models solve the task; the encoding effect is small, but measured where it is compressed.

arm	encoding	position	accuracy
MapPoPE-Flat	PoPE	path-integrated	0.994 ± 0.017
VanillaEM_P0 (MapEM-os, single p_0)	–	path-integrated	0.987 ± 0.009
Vanilla (MapWM)	RoPE	path-integrated	0.967 ± 0.039
PlainFlat	RoPE	index	0.534 ± 0.040
RoPE	RoPE	index	0.530 ± 0.043
PoPE-Flat	PoPE	index	0.509 ± 0.001
position main effect			+0.461
encoding main effect (MDE 0.029, 5/8)			+0.003

5.1 The paper task: position by encoding

Table 8 crosses the encoding (RoPE or PoPE) with the source of the angle (the index, or MapFormer’s path integral), with parameters matched within 0.4%. Index models sit just above the 0.506 blank floor on a fresh map; path-integrated models solve the task. The position main effect is +0.461; the encoding main effect is +0.003, inside an MDE of 0.029 (5/8 seeds), which is small against the position effect. It is measured where index arms sit near the floor (0.509–0.534) and path-integrated arms near ceiling (0.967–0.994), where an encoding effect is compressed. Under the converged recipe the encoding main effect is itself detectable beyond training length (Table 9); a separate torus batch (grid 64, torus recipe) agrees, with PoPE’s magnitude on top of path integration detectable beyond training length (+0.037, MDE 0.032, at $T = 512$; +0.101, MDE 0.065, at $T = 1024$; Section 11), and on MiniGrid with commanded actions the encoding is the largest main effect (+0.076, no sd recorded; Section 6.3). The converged factorial (Table 9) settles the paper-task comparison and is the headline measurement. No results file reports an sd or MDE for the paper-recipe position main effect of Table 8; its per-arm sds (at most 0.043) show it is far outside seed noise. That table was trained with the paper’s recipe (16 epochs, linear decay from step one), under which index models are known to be undertrained (Section 5.4); it is the paper-recipe measurement.

Under the converged recipe. A pre-registered rerun of this factorial under the torus recipe, with $r = 4$ path-integrated arms added (PAPER2X2_PREREG.md; 6 arms $\times$ 8 seeds, one batch), is Table 9. All three registered predictions held.

- **H1.** The position main effect at $r = 2$ and training length is +0.243 (MDE 0.038, 8/8), above the pre-registered +0.15, so the headline stands. Under the paper’s recipe, in a different batch and evaluator,

Table 9: Paper task under the converged torus recipe (300 epochs, 5% warmup + cosine, lr 10^{-3}). Metric: accuracy of predicting the observation at revisited cells, on a map not seen in training; trained at $T = 128$, evaluated at $T = 128, 512$ and 1024 (eight times the training length). 6 arms $\times$ 8 seeds in one batch, pre-registered (PAPER2X2.PREREG.md). Arms cross two factors. *Enc.* (encoding): RoPE = rotation only; PoPE = PoPE’s content-dependent magnitude with a rotation. *Position* (what sets the rotation angle): index = the token’s index, as in a standard transformer; path-int. = MapFormer’s running sum of learned per-token increments, through a bottleneck of rank r . **Vanilla** = MapFormer-WM; **RoPE** = MapFormer with index rotation. Top rows: accuracy mean $\pm$ sd; final training loss as mean (range over seeds). Bottom rows: main effects on accuracy at the stated rank, paired by seed, given as effect (MDE, seeds with a positive difference). *Position* = path-integrated minus index, averaged over the two encodings; *encoding* = PoPE minus RoPE, averaged over the two position sources; the two index arms serve both ranks. MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$ on the paired differences; every main effect shown exceeds its MDE (detectable). Effects are on raw accuracy; no position effect at matched final training loss is reported, because index and path-integrated arms end training at non-overlapping losses (see text). Blank floor (always predicting blank) about 0.51 at $T = 128$ (0.506 measured on a different evaluator’s events). Reading: path integration is the larger factor at every length; the encoding is detectably negative at training length and adds a smaller, detectable amount beyond it.

arm	enc.	position	$T = 128$	$T = 512$	$T = 1024$	final loss (range)
RoPE	RoPE	index	0.805 ± 0.012	0.470 ± 0.093	0.374 ± 0.135	0.770 (0.676–0.831)
PoPE-Flat	PoPE	index	0.679 ± 0.058	0.639 ± 0.046	0.607 ± 0.033	0.815 (0.728–0.957)
Vanilla	RoPE	path-int., $r = 2$	0.971 ± 0.043	0.915 ± 0.052	0.777 ± 0.094	0.089 (0.002–0.384)
MapPoPE-Flat	PoPE	path-int., $r = 2$	0.999 ± 0.001	0.974 ± 0.011	0.921 ± 0.023	0.006 (0.000–0.023)
Vanilla_r4	RoPE	path-int., $r = 4$	1.000 ± 0.000	0.985 ± 0.006	0.928 ± 0.012	0.0002 (0.0000–0.0003)
MapPoPE_r4	PoPE	path-int., $r = 4$	1.000 ± 0.001	0.982 ± 0.018	0.941 ± 0.028	0.0002 (0.0001–0.0005)
position, $r = 2$			+0.243 (0.038, 8/8)	+0.390 (0.079, 8/8)	+0.359 (0.106, 8/8)	
encoding, $r = 2$			−0.049 (0.038, 1/8)	+0.114 (0.030, 8/8)	+0.189 (0.061, 8/8)	
position, $r = 4$			+0.258 (0.027, 8/8)	+0.429 (0.061, 8/8)	+0.444 (0.074, 8/8)	
encoding, $r = 4$			−0.063 (0.031, 0/8)	+0.083 (0.039, 8/8)	+0.123 (0.062, 8/8)	

the same contrast was +0.461 (no MDE recorded); both index arms score higher under the converged recipe (RoPE 0.530 $\rightarrow$ 0.805, PoPE-Flat 0.509 $\rightarrow$ 0.679).

- **H2.** Registered at $r = 2$, $T = 128$ against $T = 1024$: +0.243 $\rightarrow$ +0.359, both detectable, 8/8. It is +0.390 at $T = 512$ in between. At $r = 4$ (not registered) it is +0.258, +0.429 and +0.444, 8/8.
- **H3.** The encoding main effect at $r = 2$ and training length is −0.049 (MDE 0.038), inside the registered 0.05 bound by 0.001; the estimate is itself detectable against zero (1/8 seeds positive). The registered falsifier, a detectable encoding effect larger than the position effect at any length, did not fire. Beyond training length the encoding is detectably positive: +0.114 and +0.189 at $r = 2$, +0.083 and +0.123 at $r = 4$.

Not registered: at $r = 4$ the interaction is detectably negative beyond training length (−0.172, MDE 0.074, at $T = 512$; −0.221, MDE 0.117, at $T = 1024$), where the path-integrated arms are near ceiling; at $r = 2$ it is unmeasured there, and at training length it is detectably positive at both ranks (+0.154, +0.125). Both $r = 4$ path-integrated arms converge on every seed and score 1.000 at training length, while $r = 2$ **Vanilla** reaches a final loss of 0.38 on its worst seed. Seed 0 of RoPE and **Vanilla_r4** is bitwise identical to the sign-ablation batch (weights and loss curve), and seeds 0–7 of both arms give identical accuracies there, so for these two arms the two tables are not independent (the sign ablation’s RoPE 0.799 is seeds 0–11 of the same deterministic runs). This is a determinism check, not a replication.

The maps are used. Destroying the context on trained models drops accuracy below the blank floor ($n = 3$, exploratory by n ; the action-stream drops, 0.604 to 0.811, far exceed every per-arm sd, at most 0.0449). **Vanilla** falls from 0.9889 intact to 0.2314 with actions shuffled, 0.1783 with actions resampled from another episode, and 0.2915 and 0.3092 with observations shuffled or resampled; **VanillaEM_P0** falls from 0.9873 to between 0.2680 and 0.3831. Likelihood becomes worse than uniform: NLL 3.68–4.94 nats with actions resampled against 0.006–0.100 intact, where uniform is $\ln 21 = 3.04$. The models are overconfident, not hedging. EM retains more than WM under action resampling (0.362 against 0.178, 3/3); no mechanism is claimed. This shows the models use the action stream; that the task is not solved without path integration is what the index arms of Tables 8 and 9 show (0.805 at best, against 0.971–1.000).

Why index models exceed the floor at all. Under the paper’s recipe, binned by the number of steps since a cell was last visited, index models exceed the per-bucket blank rate only at interval 1–2, which is

Table 10: Parity (chance 0.5): accuracy of path-integrated **Vanilla** (MapFormer-WM) minus index **RoPE** (same architecture, position = token index), trained at $L = 16$, evaluated at $L = 16$ –256; $n = 8$ per arm, one batch. Difference = mean paired difference; MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$; seeds + = seeds with a positive difference. Reading: path integration beats an index code detectably at every length on 8/8 seeds, peaking at $L = 32$; a replication.

	$L = 16$	$L = 32$	$L = 64$	$L = 128$	$L = 256$
difference	+0.316	+0.326	+0.167	+0.083	+0.041
MDE	0.165	0.078	0.036	0.020	0.011
seeds +	8/8	8/8	8/8	8/8	8/8

an out-and-back retrace: **RoPE** 0.557 and **PlainFlat** 0.575 against a blank rate of 0.504. From interval 3 onwards they sit on the floor, while **Vanilla** holds 0.881 at intervals of 65 or more. Under the paper’s recipe an index code learns only local retraces from action tokens used as content (Appendix A, Table 34); under the converged recipe it reaches 0.805 (Table 9), which retraces alone cannot produce. The seed count of this analysis is not recorded; it is exploratory.

5.2 Blind continuation

Match-Query withholds observations in the query phase, which closes the retrace route. On the 64^2 map, path-integrated **MapWM-Flat** scores 0.730 ± 0.247 ($n = 5$) and index **PlainFlat** 0.154 ± 0.018 ($n = 5$; seeds pooled across two runs, compared unpaired), with no overlap: the worst path-integrated seed (0.398) is above the best index seed (0.178). Against the corrected never-moved floor of 0.0893, the index model is barely above it. Context destruction ($n = 3$) collapses the path-integrated model from 0.918 to 0.074 (explore observations shuffled) and 0.076 (query actions shuffled).

On the 128^2 map, under a warmup-plus-cosine recipe, with eight seeds per arm pooled from three launches, a one-layer path-integrated model scores 0.456 ± 0.220 and a parameter-matched index model 0.108 ± 0.025 , against chance 0.0625. The paired estimate is +0.348 (MDE 0.215, 8/8), but same-seed retrains on this task drift by 0.185 per seed, so pairing is unreliable and no unpaired test of this contrast was run. Blind queries remain answerable well beyond training length: on three seeds of the 64^2 model the path-integrated arm scores 0.904, 0.894, 0.831 and 0.693 at query lengths 256, 512, 1024 and 2048, against 0.150–0.093 for the index arm (exploratory, Appendix A).

5.3 Beyond navigation: parity and contextual counting

Parity. A signed prefix sum taken modulo 2π is a parity register. Trained at $L = 16$ with eight seeds per arm in one batch, path integration beats an index code at every evaluation length (Table 10). The effect peaks at $L = 32$ and shrinks beyond it, but stays detectable on 8/8 seeds. The copy task, run as a control, failed as a control: it had no dynamic range at any length (1.000 at $L = 16$, chance from $L = 32$), so it cannot rule out a pipeline artefact. That signed rotations solve parity in a single layer is Selective RoPE’s Section 4.2 and [Grazzi et al. \(2025\)](#); this is a replication.

Recency. On k -back retrieval with uncounted filler, index arms fail and path-integrated arms succeed (Table 18 in Section 7.3). Path-integrated minus index is +0.750 (MDE 0.030, 8/8) at $T = 1024$ and +0.704 (MDE 0.056, 8/8) at $T = 2048$. The index arms appear to be at a capability limit, not undertrained: doubling the budget moves RoPE from 0.251 to 0.242 and PlainFlat from 0.247 to 0.248 (3 seeds, exploratory). Per offset, index arms score 0.99–1.00 at $k = 1$, 0.27–0.33 at $k = 8$ and 0.11–0.13 at $k = 64$; path-integrated arms are flat in k . That a fixed index code cannot count contextually is CoPE’s published claim; this is a replication.

5.4 The recipe objection

The strongest objection to Table 8 is that its index baselines are undertrained. Under the torus recipe (300 epochs, warmup and cosine, lr 10^{-3}), RoPE reaches 0.799 ± 0.018 at $T = 128$, still at final loss 0.7844 against the signed path-integrated arm’s 0.0002 (Table 15). At that recipe, signed path integration minus RoPE,

loss-matched, is -0.021 (MDE 0.013) at $T = 128$, $+0.123$ (MDE 0.075) at $T = 512$ and $+0.195$ (MDE 0.107) at $T = 1024$, negative on 0/12 seeds at the two longer lengths.

The converged factorial answers the objection on raw accuracy. Trained under the converged recipe, the index arms score higher and the position main effect is $+0.243$ (MDE 0.038, 8/8, Table 9), against $+0.461$ under the paper’s recipe in a different batch and evaluator. The position effect is reported on raw accuracy only, and not at matched loss. Matching would normally separate a model that represents the task better from one that merely trained further, but here index arms end training at loss 0.68–0.96 and path-integrated arms at 0.00–0.38, because an index code cannot fit the task; matching on loss would remove the very difference being measured. Whether path integration represents the map, rather than only fitting it, rests on the interventions of Sections 6–8. The encoding effect, whose arms do share a loss range, survives loss-matching at every length. In the sign ablation’s batch, loss-matching at $T = 128$, where $r(\text{loss}, \text{accuracy}) = -0.978$, also partials out the very deficit the index code has, so the $T = 128$ loss-matched value is not evidence that the index code maps. On Match-Query, under warmup and cosine decay at $\text{lr } 3 \times 10^{-4}$ for 300 epochs, the index arm stays at 0.108, because the blind query removes the retrace route, and on recency doubling the budget moves nothing (Section 5.3). On a small family tree, path integration beats an index code by $+0.205$ (MDE 0.118, 3/3, exploratory; Appendix C).

A second objection is that a plain transformer path-integrates through attention when actions are in its input. It does, to a degree: on the compositional task a plain index transformer transfers at 0.216 against a floor of about 0.072 (Appendix D), and on MiniGrid with commanded actions the index arms are the best arms (Section 6.3). That boundary is Claim 2.

6 Claim 2: the environment must make displacement a function of the token

MapFormer adds a fixed increment per token. Under turn-and-forward actions the displacement produced by “forward” depends on the accumulated heading, which no fixed increment can represent. On the torus the position effect falls from $+0.438$ to $+0.050$ under rotation-based actions, and recording the realised displacement instead restores it to $+0.488$ ($n = 8$). On MiniGrid with allocentric actions, path integration beats an index code only at $r = 4$ ($+0.034$, MDE 0.014, 8/8). On MiniWorld the effect does not shrink as observation aliasing falls: in converged cells it is $+0.178$ at 32 cells per token ($n = 5$, 400 epochs) and $+0.305$ at 2 ($n = 3$, 800 epochs, exploratory).

6.1 Rotation actions and allocentric recording

Five properties differ between the torus and MiniGrid. Turning each on separately in the torus code (three seeds per knob, Appendix E) singled out rotation-based actions, which were then re-run at eight seeds together with the baseline, all five knobs combined, and an allocentric record (Table 11, Figure 1). Under the allocentric record the agent’s dynamics are byte-identical to the rotation condition; only the recorded action token changes, from the commanded turn or forward to the realised absolute displacement (or STAY). The answer-stream n -gram gates are identical to three decimals under both records (orders 1/2/3/5: 0.501/0.472/0.440/0.462 against a marginal of 0.507), and the index arm stays at the floor under both.

Rotation alone takes the position effect from $+0.438$ to $+0.050$, a reduction of 0.388; all five knobs combined take it to -0.076 , a change of -0.514 . The mechanism is supported by an intervention on the action record alone: the same dynamics, recorded so that each token names a fixed displacement, restore the effect to $+0.488$ against the baseline’s $+0.438$ (no MDE recorded; trained at matched supervised events, four times the baseline’s batches). The recoding supplies an integrable input, not the target, but it removes the heading-to-displacement computation from what is learned: the result shows that a fixed per-token increment cannot perform that computation, not that a map can do without it. It requires the agent’s heading to be known, which holds in every simulator used here.

Table 11: Torus knob sweep: accuracy predicting the observation at revisited cells of a held-out map, trained and evaluated at $T = 128$. Rotate = turn and forward actions (displacement depends on heading); allocentric record = same dynamics, each action token the realised displacement; combined = rotation plus four other MiniGrid-like properties. Path-integrated = **Vanilla** (MapFormer-WM); index = **RoPE** (position = token index). Mean $\pm$ sd, $n = 8$ per arm; effect = difference of means; floor = measured no-model baseline. All arms trained 16 epochs; rotate and allocentric at 392 batches per epoch (matched supervised events, since only moves are scored), baseline and combined at 98. No sd or MDE was recorded for the effects. The combined condition’s order-3 answer-stream gate reached 0.634 against a 0.536 marginal ($n = 3$ gate), so its effect is approximate. Reading: rotation removes most of the effect and the allocentric record restores it, with the index arm at the floor under both.

condition	floor	path-integrated	index	effect
baseline (translation actions)	0.506	0.967 ± 0.039	0.529 ± 0.044	+0.438
rotate (turn / forward)	0.508	0.558 ± 0.026	0.508 ± 0.006	+0.050
rotate, allocentric record	0.508	0.996 ± 0.005	0.508 ± 0.006	+0.488
all five knobs combined	0.552	0.708 ± 0.068	0.785 ± 0.022	−0.076

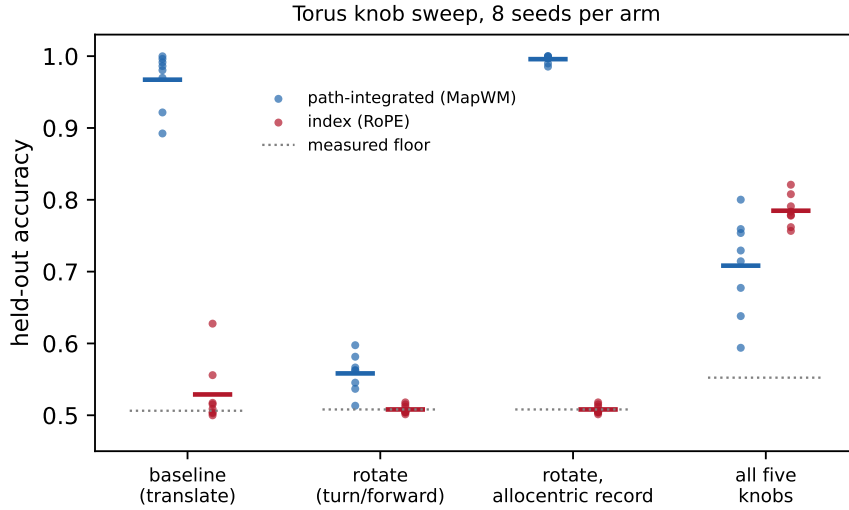


Figure 1: Per-seed accuracy for Table 11. Dotted lines are each condition’s measured floor; bars are arm means. Generated by `figures/make_knob_sweep.py` from `KNOB_SWEEP_n8.json`.

6.2 Heading resolution

At 12 headings, with real-valued position, the allocentric record still separates the arms: every path-integrated seed is above every index seed at every budget tried, the weakest path-integrated seed scoring 0.661 and the strongest index seed 0.555 (floor 0.508–0.509; $n = 3$ per budget, three budgets). The size is not monotone in budget: +0.264 at 980 batches, +0.383 at 2000 and +0.286 at 4000, where two of three path-integrated seeds converged to worse losses and the index arm left the floor. Accuracy tracks final training loss at $r = -0.996$ over the 18 runs, so arm-to-arm variation here is convergence. The direction is an existence statement; the magnitude is exploratory.

6.3 An external benchmark: MiniGrid

Commanded actions. On MiniGrid DoorKey-16x16 with egocentric observations and the environment’s own turn-and-forward actions, the full $2 \times 2 \times 2$ factorial (encoding, position, hierarchy; three layers, parameter-matched, $n = 8$) puts index arms first: **PoPE-Hier** 0.955 ± 0.003 and **PoPE-Flat** 0.953 ± 0.003 at $T = 1024$, with the paper’s **MapWM-Flat** last at 0.823 ± 0.088 (floor 0.490). The main effects at $T = 1024$ are encoding +0.076, hierarchy +0.048 and position −0.021; no sd was recorded for them, so they are directional. The arm means are citable levels. This is the regime Section 6.1 predicts: commanded rotation actions make a fixed per-token increment misspecified.

Table 12: Allocentric MiniGrid DoorKey-16x16 (actions recorded as realised displacement): held-out observation-prediction accuracy at $T = 1024$, trained at $T = 128$; $n = 8$ per arm, one batch, floor 0.490. **RoPE** = index position; **Vanilla** = MapFormer-WM; **VanillaEM** = MapFormer-EM, separate origins (discussed in Section 8.2); **r4** = bottleneck rank 4, else 2. Parameters 613,576–614,664 across the five arms (training logs). Contrast = arm minus RoPE, paired by seed: delta (MDE, seeds with a positive difference); MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$; detectable = beyond it; unmeasured = within it, not evidence of zero. Reading: MapWM beats the index arm on every seed only at $r = 4$, by a small amount.

arm	accuracy	worst seed	contrast with RoPE
RoPE (index)	0.788 ± 0.019	0.754	–
Vanilla ($r = 2$)	0.778 ± 0.045	0.693	–0.010 (MDE 0.056, 4/8), unmeasured
Vanilla_r4	0.822 ± 0.015	0.798	+0.034 (MDE 0.014, 8/8), detectable
VanillaEM ($r = 2$)	0.763 ± 0.081	0.572	–0.026 (MDE 0.087), unmeasured
VanillaEM_r4	0.803 ± 0.056	0.666	+0.015 (MDE 0.062, 7/8), unmeasured

Allocentric actions. With actions recorded as realised grid displacement (Table 12; one batch, $n = 8$, 50 epochs), MapWM at $r = 4$ beats the index arm on every seed: **Vanilla_r4** minus RoPE is +0.034 (MDE 0.014, 8/8) at $T = 1024$. At the paper’s $r = 2$ the contrast is –0.010 (MDE 0.056, 4/8), unmeasured. The effect is small, on a benchmark where every arm is far above the floor; the claim is the sign conditioned on $r = 4$, not its size. An earlier three-seed factorial found the same sign flip at $r = 2$ with effects of about 0.02, compared across batches (Appendix E).

6.4 MiniWorld: convergence, aliasing and map extent

MiniWorld’s OneRoom environment is used as a continuous 3D geometry generator: location-keyed aliased tokens are emitted on a fresh map per episode, and actions are recoded to the exact cell transition.

Convergence first. At 100 epochs with linear decay, RoPE never converged at grid 16 or above (0/9) and Vanilla never at grid 8 (0/3), so an apparent crossover in grid size tracked which arm had converged. At 400 epochs with warmup and cosine every arm is flat. There, path integration is ahead of the index code at grid 32 (+0.173, $n = 3$; extended below to $n = 5$, where it is detectable) and both arms solve grid 8 (–0.010, sd 0.007, with RoPE at 1.000 on all three seeds). No regime was found in which the index code is genuinely better.

Aliasing does not drive the effect. Holding the map at grid 32 and varying only the number of observation types changes nothing but the labelling: label mass (50.4 scored events per trajectory) and median revisit lag (33) are identical across conditions. The pre-registered aliasing hypothesis predicted that the effect would collapse as aliasing fell. At 400 epochs every level meets outcome B’s numeric condition (effect above 0.150), but the $n_{\text{obs}} = 64$ and 256 arms were not all converged and the scripted verdict was withheld (“not all arms converged”, Table 13). The falsification therefore rests on the converged cells: +0.178 at $n_{\text{obs}} = 16$ ($n = 5$, MDE 0.118, all arms flat) and, extended to 800 epochs, +0.305 at $n_{\text{obs}} = 256$ (sd 0.048, MDE 0.077, 3/3, exploratory) with every arm flat. No test of the difference between these two was run, and they differ in budget and n . The post-hoc pooled reading of the next paragraph points the same way. The final losses of the two arms do not overlap at $n_{\text{obs}} = 64$ and 256, so this study measures “optimises better”, not “represents better at equal fit”.

Map extent, an open threshold. At matched aliasing (2 cells per token), pooling the conditions available after the fact gives the pattern in Table 14: flat at 32 and 128 occupied cells, and large at 512. Within a map size the effect moves by 0.005 (grid 16) and 0.030 (grid 32) as the number of prior visits per scored cell changes; across map sizes it moves by 0.285. One hypothesis is falsified outright: at a matched number of distinct cells visited, grid 32 at $T = 128$ gives +0.275 where grid 8 at $T = 512$ gives –0.010. The map-extent reading is a post-hoc pooled analysis at $n = 3$ per point; the pre-registered pairwise verdict was that prior visits and map extent could not be separated. The two endpoints were trained at different budgets (800 and 400 epochs), and the grid-16 point is a ceiling (the index arm scores 0.971–0.994). The threshold between 128 and 512 occupied cells is therefore a boundary to test, not a finding. The MiniWorld run-to-run noise floor of 0.150 was measured on mostly unconverged runs; both it and the paired MDE are given.

Table 13: MiniWorld at fixed grid 32 (about 512 occupied cells), $T = 512$, only the number of observation types n_{obs} varied: non-blank accuracy (scored cells whose answer is not blank) on a fresh map, path-integrated (MapFormer-WM) minus index, paired by seed. Floors are the non-blank marginals; n = seeds per arm; sd = of paired differences; MDE = minimum detectable effect, $2.8 \text{sd} / \sqrt{n}$; flat = converged. Rows 1–3 are the 400-epoch study (scripted verdict withheld: not all arms converged); row 4 extends $n_{\text{obs}} = 256$ to 800 epochs. Reading: the effect does not shrink as aliasing falls; this rests on the two converged cells, which differ in budget and n , untested against each other.

n_{obs}	cells per token	floor	epochs	n	effect	sd / MDE
16	32	0.077	400	5	+0.178	0.094 / 0.118; all arms flat
64	8	0.031	400	4	+0.310	0.031 / 0.043; not all flat
256	2	0.013	400	5	+0.374	0.046 / 0.057; not all flat
256	2	0.013	800	3	+0.305	0.048 / 0.077; all flat (exploratory, $n = 3$)

Table 14: MiniWorld map extent at 2 cells per token (grid 8 at $n_{\text{obs}} = 16$ is 2 cells per token as well): non-blank accuracy on a fresh map, path-integrated minus index, paired by seed. T = episode length; prior visits = mean earlier visits per scored cell; notes give index-arm accuracy and sd / MDE (minimum detectable effect) where recorded. $n = 3$ per row, post-hoc pooling of runs at different budgets. Exploratory. Reading: flat at 32 and 128 occupied cells, large at 512; the threshold is a boundary to test, not a finding.

grid	occupied cells	T	prior visits	effect	note
8	32	512	8.64	−0.010	both arms solve
16	128	512	4.61	+0.015	sd 0.013, MDE 0.020; index 0.971–0.994 (ceiling)
16	128	1024	6.20	+0.010	index 0.987
32	512	128	1.95	+0.275	index 0.674
32	512	512	3.05	+0.305	sd 0.048, MDE 0.077; 800 epochs

6.5 What is not modelled

In Habitat, measured with the simulator but with no model trained, 69–91% of forward moves across three scenes do not produce the commanded 0.25 m displacement: the navigation mesh slides or blocks the agent. Displacement there is continuous in magnitude as well as direction, which no experiment here models.

7 Claim 3: the increment must cancel, and cancel cleanly

For the interval sum to measure net displacement, opposite actions must produce cancelling increments, and the basis they live in must be well conditioned. Forcing the increment non-negative costs MapWM 0.363 raw (MDE 0.025) and 0.280 at matched loss (MDE 0.061) on the torus at $T = 1024$ (12/12), and at matched loss leaves no measurable gain over an index code (raw, the monotone arm stays ahead of it, +0.213, MDE 0.097); Selective RoPE’s generator pays the same raw cost (−0.355 against MapWM’s −0.363). The paper’s $r = 2$ bottleneck learns a skewed basis, and $r = 4$ adds +0.085 at $T = 1024$ (8/8) for 384 parameters. The value of both depends on the task: on a counting task the same sign constraint costs MapWM an unmeasured −0.060 and Selective RoPE’s generator a small, detectable −0.068 (both raw); for that generator, its raw torus cost minus its recency cost is detectable (−0.287). In the same batch single-origin MapEM pays −0.198 raw on the counting task (MDE 0.155), detectable (−0.043 at matched loss, MDE 0.051, unmeasured), and the difference between MapEM’s and MapWM’s costs is unmeasured (−0.138, MDE 0.181). For the two WM-type generators the constraint’s cost is large on a map task and small on a counting task.

7.1 Sign

Design. Six arms, twelve seeds, one batch, torus recipe. The signed arm uses $\Delta = W_{\text{out}} W_{\text{in}} x$; **Abs_r4** uses $|\Delta|$, the one-operation isolation; **Pos_r4** and **CARoPE_r4** use the non-negative parameterisations of GRAPE’s additive path and CARoPE; **Vanilla_r4** is a construction control; **RoPE** is the index code. All five MapFormer arms have 204,757 parameters. The constraint holds after training (no negative increment in the constrained checkpoints probed, seeds 0–2 of 12 per arm), and the construction control differs from the signed arm by +0.000, +0.006 and +0.006 at the three lengths, unmeasured throughout. Rule 9 gives $r = -0.978$, −0.856 and −0.729 at $T = 128$, 512 and 1024 over 72 runs.

Table 15: Sign ablation, torus: accuracy predicting the observation at revisited cells of a held-out map, trained at $T = 128$; mean $\pm$ sd, $n = 12$ per arm, one batch, floor 0.506; final loss = arm mean. All but RoPE are MapFormer-WM at $r = 4$ with 204,757 parameters: **Signed_r4** = free-sign increment Δ ; **Vanilla_r4** = construction control; **Abs_r4** = $|\Delta|$; **Pos_r4** = softplus increment (GRAPE-style); **CARoPE_r4** = CARoPE’s increment in $(0, 1)$. RoPE = index position. Reading: at training length the signed arm is at ceiling and the constraint’s cost shows in the loss; accuracy contrasts are in Table 16.

arm	$T = 128$	$T = 512$	$T = 1024$	final loss
Signed_r4	1.000 ± 0.000	0.978 ± 0.014	0.922 ± 0.027	0.0002
Vanilla_r4	1.000 ± 0.000	0.984 ± 0.009	0.927 ± 0.015	0.0002
Abs_r4	0.946 ± 0.070	0.675 ± 0.040	0.558 ± 0.027	0.1708
Pos_r4	0.977 ± 0.035	0.798 ± 0.080	0.584 ± 0.080	0.0676
CARoPE_r4	0.900 ± 0.138	0.809 ± 0.133	0.645 ± 0.090	0.2929
RoPE	0.799 ± 0.018	0.449 ± 0.083	0.345 ± 0.118	0.7844

Table 16: Sign ablation contrasts on the torus (arms as in Table 15: **Signed** free sign; **Abs**, **Pos**, **CARoPE** non-negative; **RoPE** index), first-named minus second, paired by seed, $n = 12$, trained at $T = 128$. Main value: loss-matched, i.e. compared at equal final training loss by a regression pooled over the six arms (verdicts depend on the pool); raw difference in parentheses where recorded. MDE = minimum detectable effect, $2.8 \text{sd}/\sqrt{n}$; detectable = beyond it; unmeasured = within it, not evidence of zero. Rule 9, $r(\text{loss}, \text{accuracy})$: $-0.978, -0.856, -0.729$ at the three lengths. Reading: at matched loss the monotone arm is worse than signed by 0.215 and 0.280 beyond training length and beats the index code nowhere; raw it stays above RoPE, which trains to a much higher loss.

contrast	$T = 128$	$T = 512$	$T = 1024$
Abs – Signed	-0.006 (MDE 0.018)	-0.215 (MDE 0.055; raw -0.303)	-0.280 (MDE 0.061; raw -0.363)
Pos – Signed	-0.004 (MDE 0.010), unmeasured	-0.145 , detectable	-0.305 , detectable
CARoPE – Signed	-0.017 (MDE 0.023), unmeasured	-0.018 , unmeasured	-0.134 (MDE 0.118), detectable
Signed – RoPE	-0.021 (MDE 0.013), detectable	$+0.123$ (MDE 0.075)	$+0.195$ (MDE 0.107)
Abs – RoPE	-0.028 (MDE 0.025; raw $+0.146$, MDE 0.055)	-0.092 (MDE 0.103; raw $+0.226$, MDE 0.059)	-0.085 (MDE 0.140; raw $+0.213$, MDE 0.097)

Result. Tables 15 and 16. At matched loss, the monotone arm is worse than the signed arm by 0.215 at $T = 512$ and 0.280 at $T = 1024$, on 12/12 seeds at both. At matched loss, signed path integration beats the index code by $+0.123$ and $+0.195$, and the monotone arm beats it nowhere (**Abs** – **RoPE** is unmeasured at $T = 512$ and 1024 and detectably negative at $T = 128$; so is **Signed** – **RoPE** at $T = 128$, -0.021). On raw accuracy the monotone arm is detectably above the index code at every length ($+0.146$, $+0.226$ and $+0.213$ at $T = 128, 512$ and 1024 , MDE 0.055, 0.059 and 0.097), because RoPE trains to a much higher loss. At matched loss, the measured value of a content-dependent phase over an index code on this task therefore requires the sign.

At $T = 128$ the accuracy discriminator could not show a deficit: the signed arm sits at 1.000 ± 0.000 . The pre-registered hypothesis asked for a training-length accuracy deficit in that ceiling cell, a design error recorded before any result was read. The training-length effect is in the loss: every constrained arm has a higher final training loss than the signed arm on 12/12 seeds (**Abs** $+0.1706$, **Pos** $+0.0674$, **CARoPE** $+0.2927$). The **Pos** and **CARoPE** arms carry an initialisation confound (they start close to RoPE, the signed arms close to NoPE).

The learned code. The opposition score $\|D(+a) + D(-a)\|/\|D\|$ is 0 when opposite actions cancel and 2 when they are identical. It is 0.125 and 0.106 (two axes; seeds 0–2) for the signed arm and 0.128 and 0.130 for the construction control, against 1.849/1.855 for **Abs**, 1.905/1.885 for **Pos** and 1.981/1.977 for **CARoPE**. A monotone code cannot reach 0 by construction; the informative part is that the signed arm does reach about 0.1. Cancellation chooses what the accumulator measures: signed increments give net displacement, a map; monotone increments give path length, a clock.

A second generator. A pre-registered batch (96 runs, one batch, 12 seeds per arm) repeated the constraint on Selective RoPE’s generator (**SRoPEGen**: gate, causal convolution, no rank bottleneck). On the torus, **SRoPEGen** scores 1.000, 0.984 and 0.899 at $T = 128, 512$ and 1024 and its monotone twin 0.925, 0.638 and 0.545. The raw cost at $T = 1024$ is -0.355 (MDE 0.046, 0/12), against MapWM’s stored raw cost of -0.363 (MDE 0.025); the difference between the two generators’ costs is $+0.009$ (MDE 0.052), unmeasured. The raw sign cost on navigation is therefore not specific to MapFormer’s generator. The registered loss-matched form (Q1) was falsified as registered: its loss-matching pool contained only the two **SRoPEGen** arms, whose final

Table 17: Rank sweep on MapFormer-WM, torus: accuracy predicting the observation at revisited cells of a held-out map, trained at $T = 128$; only the bottleneck rank r varies (parameters in the first column); mean $\pm$ sd, $n = 8$ per arm, one batch, floor 0.506. Last column: minus $r = 2$ at $T = 1024$, paired by seed (t statistic where recorded; seeds with a positive difference). Reading: $r = 4$ adds +0.085 at $T = 1024$ for 384 parameters and cuts seed sd from 0.064 to 0.012; the gain is a step.

arm (parameters)	$T = 128$	$T = 512$	$T = 1024$	vs $r = 2$ at $T = 1024$
$r = 2$ (204,373)	0.993 ± 0.017	0.944 ± 0.028	0.834 ± 0.064	–
$r = 4$ (204,757)	1.000 ± 0.000	0.982 ± 0.005	0.919 ± 0.012	+0.085 (t 3.57, 8/8)
$r = 8$ (205,525)	1.000 ± 0.000	0.981 ± 0.018	0.925 ± 0.033	+0.091 (8/8)
$r = 16$ (207,061)	1.000 ± 0.000	0.972 ± 0.015	0.913 ± 0.018	+0.079 (8/8)
$r = 32$ (210,133)	1.000 ± 0.000	0.982 ± 0.010	0.928 ± 0.021	+0.095 (8/8)

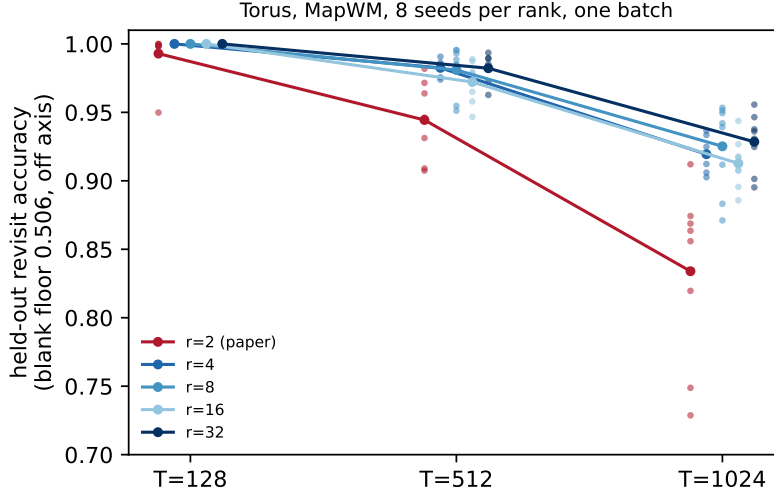


Figure 2: Per-seed held-out accuracy for Table 17; lines join arm means. Generated by `figures/make_rank_sweep.py` from `RANK_SWEEP.json`.

losses do not overlap (0.0002–0.0006 against 0.20–0.47), so the fit absorbed the effect by construction and returned -0.056 (MDE 0.083). In an exploratory eight-arm pool that includes the six stored sign arms, the loss-matched cost is -0.188 (MDE 0.059, 0/12).

Prior art. The sign axis and the parity theorem are due to Sarrof et al. (2024) and Grazzi et al. (2025), and were carried into content-dependent rotation in attention by Movahedi et al. (2026). CARoPE, GRAPE’s additive path and CoPE are non-negative. What is new here is the navigation regime and the one-operation isolation at identical parameter count.

7.2 Rank

The sweep. Only r varies across five arms, eight seeds each, one batch, torus recipe (Table 17, Figure 2). Against $r = 2$, $r = 4$ adds +0.085 at $T = 1024$ (t 3.57, 8/8) and +0.038 at $T = 512$. The gain is a step: $r = 8$, 16 and 32 give +0.091, +0.079 and +0.095 at $T = 1024$, all 8/8. $r = 4$ also cuts the $T = 1024$ seed sd from 0.064 to 0.012, which matters because sd sets every MDE. It costs 384 parameters; Selective RoPE’s gate arm (sigmoid gate plus a readout swap) reached a similar +0.086 at $T = 1024$ for 8,193 parameters, an effect that cannot be attributed to the gate (Appendix F). The pre-registered branch “accuracy rises with r ” fired. $r = 1$ and $r = 3$ were not run.

What $r = 2$ learns. Given more than two dimensions, the model still places its actions in a two-dimensional plane (plane energy 1.0000 at $r = 4$ and 0.9996 at $r = 32$), so the learned codes are two-dimensional, as the paper’s argument expects ($r = 1$ was not run). What fails at $r = 2$ is the basis: opposite actions fail to cancel

Table 18: Recency: return the k -th most recent symbol ($k_{\max} = 64$) past uncounted filler ($p_{\text{filler}} = 0.5$); retrieval accuracy on held-out episodes, trained at $T = 1024$; mean $\pm$ sd, $n = 8$ per arm, one batch; final loss = final training loss. Sign arms (MapFormer-WM, $r = 4$) as in Table 15: **Signed_r4** free sign, the others non-negative. **RoPE**, **PlainFlat** = index position. Chance 0.0625; most-recent floor 0.0771. Reading: index arms fail, path-integrated arms succeed, and the non-negative constraint’s cost is unmeasured here.

arm	final loss	$T = 1024$	$T = 2048$
Signed_r4	0.025	1.000 ± 0.000	0.940 ± 0.050
CARoPE_r4	0.020	1.000 ± 0.000	0.973 ± 0.028
Pos_r4	0.061	0.979 ± 0.041	0.947 ± 0.069
Abs_r4	0.110	0.961 ± 0.068	0.889 ± 0.126
RoPE	2.150	0.234 ± 0.025	0.234 ± 0.011
PlainFlat	2.143	0.236 ± 0.019	0.233 ± 0.009

by half the action scale (opposition 0.4950 against 0.0922 at $r = 4$), and north and east are nearly parallel ($|\cos| = 0.7833$ against 0.1754). This is a description of the learned codes on eight seeds; the causal route from skew to accuracy is not tested. The paper’s own reported limitation, non-orthogonal actions (0.779 in Table 6), falls to 0.174 at $r = 4$ with no regulariser.

Scope. The rank result is conditional.

- **MapPoPE.** $r = 4$ against $r = 2$ on MapPoPE is $+0.019$ at $T = 1024$ (5/8; loss-matched $+0.015$), unmeasured. PoPE has 64 frequency channels per head to MapWM’s 32, which may explain it; that reason is untested. “Use $r = 4$ ” is a MapWM-family recommendation.
- **Match-Query.** Rank alone adds $+0.005$ (MDE 0.091), unmeasured; with a four-pass loop it adds $+0.154$ (MDE 0.084, 8/8), paired on a task where pairing is unreliable (Section 9.1).
- **MiniGrid.** The allocentric path-integration gain exists only at $r = 4$ (Section 6.3).

Not packing geometry. If $r = 2$ ’s deficit were a packing effect of D displacement axes into r dimensions, it should grow with D , which was proposed as an account of the paper’s 5D collapse. On a D -dimensional torus ($D = 2, 3, 5$; eight seeds; gates at chance) $r = 2$ posts its best score at $D = 5$ (0.896 ± 0.076) and its deficit against the best higher rank is $+0.110$, $+0.153$ and $+0.055$ at $D = 2, 3$ and 5 at $T = 512$, not growing with D . Two of three pre-registered falsifiers fired and the account is withdrawn; the paper’s own 5D run used $r = D$ (App. D.1). That the gap is an optimisation effect rests on one independent detectable cell, $r = D + 2$ minus $r = D$ at $D = 5$ ($+0.073$, t 3.35, 8/8), and is exploratory.

7.3 The task sets the value

Recency. The same arms trained on k -back retrieval with uncounted filler (Table 18; eight seeds, one batch) give a different picture. $T = 1024$ is at ceiling for two arms, so the registered comparison is at $T = 2048$. There, **Abs** minus **Signed** is -0.052 raw (MDE 0.105) and $+0.016$ loss-matched (MDE 0.046), and the mean of the three monotone arms minus **Signed**, the registered primary, is -0.004 raw (MDE 0.055) and $+0.026$ loss-matched (MDE 0.027): all unmeasured.

The pre-registered MONOTONE batch measured the recency half again, in one batch with the torus half for the second generator. On recency at $T = 1024$, the monotone constraint costs MapWM -0.060 (MDE 0.084, 1/12), unmeasured, and costs **SROPEGen** -0.068 (MDE 0.054, 2/12), a small detectable amount; at $T = 2048$, in contrasts not registered (exploratory), both are unmeasured (-0.015 , MDE 0.097; -0.052 , MDE 0.076). For **SROPEGen**, the torus cost minus the recency cost is -0.287 (se 0.025, MDE 0.071), detectable. The accurate form of the crossover is therefore that, for the two WM-type generators, the sign constraint is large on a map task and small, but not always unmeasured, on a counting task; single-origin MapEM pays a detectable -0.198 there (Section 8.5).

What the free arm learns. The unconstrained arm learns a different accumulator on each task. The growth exponent α of the accumulator’s range, $\text{range}(S) \sim T^\alpha$, is 0.591 ± 0.028 on the torus and 0.967 ± 0.009

Table 19: Recency gate ablation: eval-only interventions on the increments Δ of trained **Signed_r4** (MapFormer-WM), $T = 1024$, $n = 8$; accuracy is the seed mean ($\pm$ sd where recorded). Content tokens are counted, filler is not. Chance 0.0625; index arms score 0.234. **scale_match** is the control for **equalize**; the **uniform** pair is magnitude-matched and not pre-registered. Reading: a content gate is most of the counting mechanism (+0.594, 8/8), and **equalize** is confounded.

condition	what it does	accuracy
none	—	1.0000
zero_filler	filler increments set to zero	0.9997
equalize	filler increments set to the mean content increment	0.0855
scale_match	only the scale of θ changed (control)	0.1096
uniform_content	magnitude-matched constant on content tokens	0.7826 ± 0.1150
uniform_all	the same constant on every token	0.1886 ± 0.0248
zero_content	content increments set to zero	0.0680
zero_query	query increments set to zero	0.8232 ± 0.0797

on recency for **Signed_r4**, while the constrained arms sit near 1 on both (**Abs_r4** 1.010 and 0.976).

How it counts, by intervention. Eval-only interventions on the trained signed arm (Table 19) show that a content gate is most of the counting mechanism. Removing filler increments costs -0.0003 . Replacing Δ with a magnitude-matched constant on content tokens only scores 0.7826; the same constant on every token scores 0.1886: +0.594 on 8/8 seeds. An “equalize” condition that looked decisive (0.0855) is confounded, because a control that changes only the scale of θ collapses as hard (0.1096); the magnitude-matched pair was added after that control failed and is labelled as not pre-registered. Gate strength does not predict accuracy near ceiling ($r = -0.34$). That position should count only selected tokens is CoPE’s mechanism; that an unconstrained MapFormer discovers such a gate is the new observation.

Recency does not need a clock. A signed accumulator can do k -back without counting upward. If symbols carry $\Delta = (1, 0)$ and the query token for offset k carries $\Delta = -(k - 1, 0)$, the prefix sum rewinds the count at the query and the retrieval offset is zero for every query; a single-origin kernel then selects the answer on 1423/1423, 1417/1417 and 1415/1415 queries in three draws. Section 8 shows the full model can hold this solution. The recency half of the crossover shows that for MapWM a monotone increment is cheap there, not that it is required.

7.4 One description, two severities

Across four torus arms from two batches, α is 0.518 (**Signed_r4**), 0.524 (**Vanilla_r4**), 0.619 ($r = 2$) and 0.943 (**Abs_r4**), and $r(\text{opposition}, \alpha) = +0.9995$. Sign and rank are then one finding at two severities: the monotone code cannot cancel at all, the $r = 2$ code cancels badly. This is exploratory (four points, correlational), and α is a statistic fitted to trained models, nearly collinear with the simpler opposition score, not a parameter that can be varied. Two analyses give different torus exponents for **Signed_r4** (0.518 here, 0.591 in the recency comparison); each is quoted only beside its own comparison.

The route from a growing accumulator to failure beyond training length is not established. The one account imported for it, that low-frequency channels never complete a cycle at training length and are read at unseen phases later, predicts that ablating those channels costs less out of distribution. It costs more: ablating the 32 lowest-frequency of its 64 channels ($2 \text{ heads} \times 32$) of **Signed_r4** costs -0.001 at $T = 128$ and -0.139 ± 0.092 at $T = 1024$ (Section 11).

8 Claim 4: a shared kernel is representable but harder to find

MapEM factorises where from what completely: one position kernel is shared by every query-key pair. On map tasks no detectable difference from MapWM was found once its initialisation is fixed, including beyond the training length on MiniGrid. On varying-offset recency, single-origin MapEM trails MapWM by 0.375 (MDE 0.154, 0/8). The gap is not in what the model can represent: the solution installed and frozen scores 1.000 on 8/8 seeds, and installed at training scale with a content leak closed it is held at 1.000 on 8/8. From

scratch MapEM finds the solution per query token, in wrapped form, and a single shared offset is found on 7/8 seeds. At a fixed budget, fewer offsets are better (+0.665, MDE 0.213, 8/8). Per-pair position origins add +0.215 over single-origin MapEM at $n = 48$, split between pathway capacity (+0.124) and per-pair freedom (+0.091), and only the freedom changes the mechanism (per-token rewind share $0.948 \rightarrow 0.189$). Every recency contrast in this section is a contrast of fit, with $r(\text{loss}, \text{accuracy})$ between -0.936 and -0.987 ; the “search, not representation” reading rests on the existence and holding constructions, not on loss-matching.

8.1 Two ways to compose where and what

Section 2 set out the algebra: MapEM’s $A_X \odot A_P$ is TEM’s conjunction with one kernel for every pair, up to a sign gauge; MapWM’s kernel has per-pair amplitudes and phases. A probe of the circular standard deviation of the kernel phase across 29,056 query–key pairs per model confirms the parameterisation in trained recency models: 1.947 for trained MapWM (seeds 1.83–2.08; 2.722 untrained), exactly 0.000 for both MapEM forms, against a uniform null of 3.267. Single-origin MapEM keeps 25 of 64 frequency blocks alive; separate-origin MapEM all 64. The only reading licensed is that MapWM can reshape its kernel per pair and MapEM cannot. Training reduces MapWM’s spread, so the probe does not show that MapWM’s advantage comes from reshaping.

8.2 Map tasks

No detectable difference once initialisation is fixed. On allocentric MiniGrid, single-origin MapEM at $r = 4$ minus MapWM at $r = 4$ is +0.0012 ($T = 512$, 4/8) and +0.0035 ($T = 1024$, 6/8), unmeasured (no MDE was recorded); its worst-to-second-worst seed gap is 0.002, against 0.137 for the paper’s separate-origin $r = 2$ form. On the torus vocabulary sweep at $n_{\text{obs}} = 256$, the same contrast is +0.0597 (MDE 0.169), unmeasured, all of it one collapsed `Vanilla_r4` seed; with each arm’s worst seed dropped it is +0.0000. Neither contrast is detectable.

The origin design is task-dependent. The paper’s separate q_0, k_0 is worse than a single p_0 on map tasks: on the torus recipe separate minus single is -0.154 (MDE 0.130, 0/8) at $T = 1024$, and on the paper task and Match-Query it is worse at $n = 3$ (Section 4). On varying-offset recency it is directionally better: +0.128 (MDE 0.108, 17/24) pooled over 24 seeds, but +0.073 (MDE 0.130, 9/16) on the 16 fresh seeds alone, unmeasured.

Length on the paper task. On the paper’s OOD-s condition extended to longer episodes, a pre-registered 50-epoch rerun found MapEM degrading more slowly than MapWM (Table 20). Floor-normalised, MapEM minus MapWM is +0.067 (MDE 0.168) at $l = 512$, +0.186 (MDE 0.185, 8/8) at $l = 1024$ and +0.287 (MDE 0.194, 8/8) at $l = 2048$. Rule 9 gives $r = -0.461$, so this is not simply a loss gap. The pre-registered convergence gate (every arm at $\text{IID} \geq 0.99$) failed, with MapWM at 0.968 and MapEM at 0.985, so the registered verdict is not read; the numbers are exploratory. The effect is monotone in length, the axis of Section 11, and it does not support “a shared kernel is better when the offset is fixed”: other fixed-offset tasks show no detectable difference.

8.3 The recency gap

Whittington et al. (2025) predicts that episodic-memory models learn faster than working-memory models except on N-back. On varying- k recency (eight seeds per arm, one batch, recency recipe; Table 21), MapWM reaches training loss below 0.5 on 8/8 seeds, single-origin MapEM on 0/8 and separate-origin MapEM on 1/8, so the registered prediction that EM is not faster held. Single-origin MapEM minus MapWM is -0.375 (sd 0.156, MDE 0.154, 0/8) at $T = 1024$ and -0.437 at $T = 2048$; separate-origin minus MapWM is -0.137 (MDE 0.106, 1/8). The deficit grows with k : at $k = 64$ MapWM scores 0.97, separate-origin MapEM 0.73 and single-origin MapEM 0.37. Single-origin MapEM is well above chance (0.600 against 0.0625).

The varying- k task is not Whittington et al. (2025)’s N-back, which has a fixed N and no filler; the fixed- k condition of Section 8.5 is the analogue, and there MapEM reached training loss below 0.5 in a median 54

Table 20: Paper task, extended OOD-s condition (larger map, $p_{\text{empty}} = 0.8$) at episode length l , 50-epoch rerun, $n = 8$ per arm, one batch: accuracy predicting the observation at revisited cells of a held-out map, mean $\pm$ sd (floors 0.799–0.802 at $p_{\text{empty}} = 0.8$). **Vanilla** = MapFormer-WM; **VanillaEM_P0** = MapFormer-EM, one origin; **MapPoPE-Flat** = PoPE magnitude with MapFormer’s phase. Contrasts: floor-normalised (fraction of the range above floor), paired by seed; MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$. The pre-registered convergence gate failed; exploratory. Reading: MapEM degrades more slowly with length than MapWM.

	$l = 512$	$l = 1024$	$l = 2048$
Vanilla (MapWM)	0.964 ± 0.029	0.927 ± 0.034	0.886 ± 0.035
VanillaEM_P0 (MapEM)	0.978 ± 0.015	0.964 ± 0.016	0.942 ± 0.024
MapPoPE-Flat	0.991 ± 0.003	0.978 ± 0.004	0.963 ± 0.005
MapEM – MapWM, normalised	+0.067 (MDE 0.168)	+0.186 (MDE 0.185)	+0.287 (MDE 0.194)
MapPoPE – MapEM, normalised	+0.066 (MDE 0.079)	+0.073 (MDE 0.090)	+0.102 (MDE 0.128)

Table 21: Varying- k recency (return the k -th most recent symbol): retrieval accuracy on held-out episodes, trained at $T = 1024$, mean $\pm$ sd, $n = 8$ per arm, one batch. **Vanilla_r4** = MapFormer-WM; **VanillaEM_r4** = MapFormer-EM, separate origins; **VanillaEM_P0_r4** = MapFormer-EM, one origin. Final loss = range over seeds; worst = worst seed at $T = 1024$; last column = seeds reaching training loss < 0.5 (median epoch). Chance 0.0625, most-recent floor 0.0771. Reading: at a shared budget single-origin MapEM trails MapWM by 0.375 at $T = 1024$, separate-origin by 0.137.

arm	final loss	$T = 1024$	$T = 2048$	worst	loss < 0.5
Vanilla_r4 (MapWM)	0.008–0.380	0.975 ± 0.072	0.947 ± 0.078	0.797	8/8 (median ep. 70.5)
VanillaEM_r4 (separate)	0.430–1.136	0.837 ± 0.081	0.728 ± 0.106	0.689	1/8
VanillaEM_P0_r4 (single)	0.924–1.885	0.600 ± 0.126	0.510 ± 0.111	0.365	0/8

epochs against MapWM’s 99 (descriptive; no inferential contrast). MapWM at four times the budget was not run, so -0.375 is a comparison at a shared budget and says nothing about asymptotic ability.

8.4 Existence and holdability

Section 7.3 gave a kernel-level construction that solves recency. Table 22 tests it in the full one-layer model, one rung at a time.

Existence. With the rewind installed in the position pathway and frozen, and the content branch learned from random initialisation, single-origin MapEM scores 1.000 ± 0.000 on 8/8 seeds at $T = 1024$ and $T = 2048$, at every k from 1 to 64. Frozen minus from-scratch is $+0.400$ (MDE 0.125, 8/8); frozen minus MapWM is $+0.025$ (MDE 0.071), unmeasured. The model represents the task exactly.

Holdability. Made trainable from an install written in small weights, the model falls back to 0.642. The same increment written at eight times the weight scale holds 0.941: $+0.298$ (MDE 0.270, 7/8), so most of that fall was the optimiser eroding a solution stored in tiny weights. Released at epoch 30 or 100, after the content branch had trained, the small install still broke within 1–6 and 1–19 epochs on every seed, which rules out an account in which only an untrained content branch dismantles it. Closing the content leak into Δ through W_{in} removes the residual at $8\times$ by the registered seed-count readout: with those columns held at zero, the $8\times$ install scores 1.000 ± 0.000 on 8/8 seeds (0.991 at $T = 2048$), against 4/8 seeds at ≥ 0.95 with the leak open. The paired accuracy contrast ($+0.059$, MDE 0.063) is at ceiling and unmeasured; the registered readout was the seed count. At the small scale the code is erased even with zero leak.

Scope. “Holds” is shown on one task, in one configuration, at $n = 8$, with the content columns of W_{in} pinned: a point in the weight space that the unconstrained optimiser drifts away from.

8.5 Search

Found per token, wrapped. Because θ enters attention through cosines and sines, a block’s rewind only has to hold modulo $2\pi/\omega_i$, and gradient descent reaches the nearest such point. An eval-only anatomy reads

Table 22: The existence and holdability ladder for single-origin MapEM on recency (retrieval accuracy, trained at $T = 1024$), $n = 8$ per arm (seeds 0–7). “Install” writes the rewind code (the query’s increment undoes the count, so the retrieval offset is zero) into the position pathway, frozen or trainable; “scale” is the weight scale at which it is written; “leak closed” holds the content columns of W_{in} at zero. $T = 2048$ values are means; the last column counts seeds at ≥ 0.95 , the registered readout. Reading: the solution is represented exactly, and held at $8\times$ scale with the leak closed.

arm	$T = 1024$	$T = 2048$	seeds ≥ 0.95
from scratch (VanillaEM.P0.r4)	0.600 ± 0.126	–	–
installed, frozen	1.000 ± 0.000	1.000 ± 0.000	8/8
installed at scale 1/64, trainable	0.642 ± 0.266	0.582	0/8
installed at scale 1/64, trainable, leak closed	0.784 ± 0.281	0.709	3/8
installed at scale 1/8 ($8\times$), trainable	0.941 ± 0.063	0.895	4/8
installed at $8\times$, trainable, leak closed	1.000 ± 0.000	0.991	8/8
MapWM (Vanilla.r4)	0.975 ± 0.072	–	–

Table 23: Recency with one fixed offset k (the curriculum arm introduces k gradually), $n = 8$ per arm, one batch, trained at $T = 1024$; MapEM = single-origin MapFormer-EM, MapWM = MapFormer-WM. Accuracy mean $\pm$ sd at $T = 1024$, seeds ≥ 0.9 there, mean at $T = 2048$, final training loss. Epochs are the median to training loss below 0.5 (seeds reaching it); *not comparable, since the curriculum’s first epochs ask only $k \leq 2$. Reading: with one shared offset MapEM finds the rewind on 7/8 seeds, in a median 54 against MapWM’s 99 epochs.

arm	$T = 1024$	seeds ≥ 0.9	$T = 2048$	final loss	epochs to loss < 0.5
MapEM, $k = 64$ fixed	0.985 ± 0.043	7/8	0.946	0.065	54 (8/8)
MapEM, $k = 16$ fixed	0.994 ± 0.018	8/8	0.966	0.015	25 (8/8)
MapWM, $k = 64$ fixed	0.947 ± 0.120	7/8	0.755	0.119	99 (7/8)
MapEM, k curriculum	0.727 ± 0.038	0/8	0.668	1.045	11 (8/8)*

each query token’s retrieval as a function (does the kernel select the answer?) rather than as a linear slope, across five from-scratch MapEM arms and 24 seeds each. Of the cells that fail (accuracy ≤ 0.3 , $k \geq 8$), a fraction 0.000 carry a rewind, in every arm. Of solved large- k cells, 93–97% are retrieved through the query token’s own wrapped rewind, to the kernel’s peak when the content gain A_X is positive and to its trough when it is negative: $A_X > 0$ on 1.000 of peak routes and 0.000 of trough routes in every arm, which is the sign gauge of Section 2 visible in trained models. In single-origin MapEM, 595 of 1368 large- k cells are solved, 0.427 of them through a peak rewind and 0.541 through a trough rewind. This is a description (the route table was not registered). An earlier linear-slope readout, not invariant to the $2\pi/\omega_i$ shifts, had reported no rewind in any run.

The obstacle is spread, not size. With one shared k (Table 23), single-origin MapEM finds a 63-symbol rewind on 7/8 seeds, all wrapped (none linear), in a median 54 epochs to loss below 0.5 against MapWM’s 99. At $T = 2048$ MapEM minus MapWM is $+0.191$ (MDE 0.150, 7/8), exploratory and unregistered; at $T = 1024$ it is $+0.038$ (MDE 0.133), unmeasured. This is the condition matching Whittington et al. (2025)’s N-back. A curriculum over k adds $+0.127$ (MDE 0.116, 7/8), all at $k \leq 32$ ($+0.300$ and $+0.301$ in the two bins introduced by epoch 90); tokens introduced later gain nothing measurable.

Queries per token is the currency. If each query token must find its own rewind with only the queries that name it, success should track exposure per token. With k drawn from a set of m offsets (single-origin MapEM, eight seeds per arm, one batch; Table 24), at a fixed budget $m = 4$ minus $m = 64$ is $+0.665$ (MDE 0.213, 8/8). At matched queries per token, $m = 16$ at four times the epochs minus $m = 4$ is $+0.045$ (MDE 0.166), unmeasured. Holding exposure fixed, the number of offsets is not detectably costly; holding the budget fixed, it is decisive. Exposure was varied through the budget, so queries per token and optimiser steps per token are not separated. A 300-epoch version gave $m = 4$ minus $m = 64$ of $+0.422$ (MDE 0.122, 8/8), and four times that budget took the 64-offset task from 0.578 to 0.928 on the primary readout (0.609 to 0.930 over all trained offsets; unregistered).

Availability. For each query token, the best achievable normalised kernel value in the model’s own rank-4 subspace was computed against real episodes on eight stored single-origin checkpoints. It is 0.992 for failed

Table 24: Recency with offsets k drawn from a set of size m , 60-epoch base budget, single-origin MapFormer-EM, $n = 8$ per arm, one batch, trained at $T = 1024$. Queries per token = training queries naming each query token. Primary accuracy is the mean over $k \in \{4, 16, 64\}$ at $T = 1024$, mean $\pm$ sd; final loss = final training loss. Reading: at a fixed budget fewer offsets is decisively better; at matched queries per token the difference is unmeasured (within its minimum detectable effect).

arm	queries per token	primary accuracy	final loss
$m = 4$, 60 epochs	80,600	0.913 ± 0.110	0.306
$m = 16$, 240 epochs	80,600	0.957 ± 0.098	0.295
$m = 16$, 60 epochs	20,200	0.590 ± 0.190	1.161
$m = 64$, 60 epochs	5,000	0.248 ± 0.175	2.502

Table 25: Per-pair origins on recency, $T = 1024$, $n = 48$, paired by seed; seeds 0–7 reused under a bitwise determinism check, seeds 8–47 new. **EMPair** = single-origin MapEM whose origins also read the token (per-pair kernel); **EMPairConst** = the same pathway reading a constant (kernel still shared). Delta = mean paired accuracy difference; MDE = minimum detectable effect, $2.8 \text{sd}/\sqrt{n}$; seeds + = positive differences. Rule 9: $r = -0.954$ over the 96 **EMPair** and **EMPairConst** runs (contrasts of fit). Reading: all three terms are detectable, and the freedom term replicates on fresh seeds.

term	contrast	delta	MDE	seeds +
total	EMPair – single-origin MapEM	+0.215	0.068	42/48
pathway	EMPairConst – single-origin MapEM	+0.124	0.071	34/48
freedom	EMPair – EMPairConst	+0.091	0.066	34/48
freedom, fresh seeds 8–47 only		+0.089	0.069	–

tokens (181 cells) and 0.995 for solved ones (166 cells); the value achieved is 0.359 and 0.544. A near-perfect wrapped rewind is available for tokens the model fails as much as for those it solves; at per-token grain, failures are consistent with search rather than with the absence of a solution. This is a description of eight checkpoints, not an inferential contrast. At initialisation the position pathway is nearly silent: its gradient is 1.7×10^{-4} to 4.3×10^{-4} of the content branch’s on 8/8 seeds, because the score is a product (Appendix G).

Monotone MapEM. The pre-registered MONOTONE batch asked whether MapEM needs negative increments to rewind (twelve seeds per arm, one batch; all numbers in Appendix K). Removing the sign takes single-origin MapEM from 0.641 ± 0.136 to 0.443 ± 0.117 at $T = 1024$, a cost of -0.198 (MDE 0.155, 2/12), detectable; the same constraint costs MapWM -0.060 (MDE 0.084), unmeasured; their difference, -0.138 (MDE 0.181), is unmeasured. The registered verdict is **unresolved**: “subtraction needed” required the difference to be detectable, and “wrap substitutes” required MapEM’s cost not to be. *The following reading is not a verdict.* Monotone MapEM that solves a cell still solves it through the query token’s own step in 0.936 of solved cells (0.953 signed); with non-negative increments that step is necessarily a forward, wrapped one, so the rewind route does not require subtraction. What the constraint changes is how many tokens find that step: 10.4 against 23.9 solved cells per seed (exploratory, -13.5 , MDE 11.5). A constraint that leaves the solution expressible and the route intact, and reduces how often training finds it, is what the search account predicts. Two caveats apply: rule 9 is -0.987 in this batch, and at matched loss MapEM’s cost is -0.043 (MDE 0.051), unmeasured; and the monotone route fraction averages over very few solved cells on some seeds (one seed has one solved cell, another none).

8.6 Per-pair origins

If the recency deficit is the cost of sharing one kernel, giving MapEM per-pair origins should remove the need to rewind. **EMPair_r4** sets $q_t^p = p_0 + W_{\text{out}}^q W_{\text{in}}^q x_t$ (and likewise for keys), keeping the Hadamard composition, rank and depth; W_{out} starts at zero, so at step 0 it is single-origin MapEM exactly, with 2,048 more parameters. The capacity control **EMPairConst_r4** uses the same pathway reading a learned constant, so origins are identical for every token and the kernel stays shared, with 128 more parameters than **EMPair_r4**. Both start as the same function as single-origin MapEM (maximum logit difference 0).

Decomposition. At $n = 48$ all three terms are detectable (Table 25), and the freedom term replicates on the 40 fresh seeds alone. Extra capacity in the position pathway buys about +0.12; letting those parameters

Table 26: Mechanism readouts on recency, $n = 8$ per arm, descriptive. **EMPair_r4** = single-origin MapEM whose origins also read the token (per-pair kernel); **EMPairConst_r4** = the same pathway reading a constant (shared kernel). Phase spread = circular sd of the position kernel’s phase across query–key pairs. A cell is one offset k in one model; route = fraction of solved cells (accuracy ≥ 0.9 , $k \geq 8$) retrieved through the query token’s own step, readout $\max(\text{sel} - \text{sel}_0, \text{selmin} - \text{selmin}_0) \geq 0.5$ in the best head; solved cells are per seed. Reading: only per-pair freedom abandons the rewind route and spreads the phase.

arm	phase spread across pairs	solved cells via per-token rewind	solved cells ($k \geq 8$)
single-origin MapEM	0.000	0.964	23.5
EMPairConst_r4	0.000	0.948	36.8
EMPair_r4	1.448	0.189	30.8

read the token buys about +0.09 more. At $n = 8$ the total was +0.280, so the first eight seeds overstated it. At $n = 8$, **EMPair** minus MapWM was -0.095 (MDE 0.135, 0/8), unmeasured: per-pair origins are not shown to close the gap to MapWM.

Mechanism, by intervention, on descriptive $n = 8$ readouts. The two terms change different things (Table 26). The control, with more parameters, keeps the per-token rewind route (0.948, against single-origin MapEM’s 0.964) and zero phase spread. Only per-pair freedom abandons the route (0.189) and spreads the kernel phase across pairs (1.448). Freedom, not capacity, changes the mechanism. This readout is eight seeds per arm and descriptive of those runs.

This is evidence about the varying-offset case only. It does not show that MapWM wins recency through per-pair freedom: MapWM’s trained phase spread is below its untrained spread.

8.7 Phase freedom in the origins

A smaller freedom, the n_b global phases between q_0 and k_0 , also matters. Letting the phases of k_0 move (**EMDoF_alignfree**) against a control that is the same function at initialisation, with the same optimiser scale and phases pinned (**EMDoF_magonly**), gives +0.146 (MDE 0.086, 22/24) on recency, and +0.113 (MDE 0.111, 14/16) on the 16 fresh seeds alone, which clears its MDE by 0.002. Magnitude freedom that the optimiser does move is unmeasured (-0.015 , MDE 0.083), and so is initial coherence (-0.004 , MDE 0.076). What phase freedom changes is not the route: every free-phase head peaks away from zero offset (48/48), but the peak-rewind share barely moves (0.445 against 0.425). Its gain is detectable at k 1–16 (+0.124, MDE 0.084) and 17–32 (+0.263, MDE 0.110), unmeasured beyond. The mechanism is unidentified.

8.8 Relation to the two-algorithms account

Whittington et al. (2025) derive EM’s capacity advantage at a fixed neuron budget from a separate synaptic memory network. MapEM has none: its Hadamard product is a rank-one contraction of the product space, so in that paper’s sense both MapFormers are working-memory models, and its capacity result does not transfer. Consistently, the vocabulary sweep with a healthy MapEM arm shows no detectable capacity advantage (Section 8.2). Its N-back exception corresponds to the fixed- k condition, where MapEM reached loss below 0.5 in a median 54 against 99 epochs (descriptive), not to varying- k recency.

Scope of Claim 4. One varying-offset task. Contrasts of fit throughout. Holdability at a pinned point in weight space. MapWM at the larger budget not run. On map tasks no detectable MapEM cost was found, so the claim is scoped to varying-offset retrieval.

9 Claim 5: what added machinery buys

Of the additions tested (weight-shared recursion, an explicit token gate, Selective RoPE’s generator, a forget gate, hierarchical pooling, and explicit state correction), only recursion measurably adds more than a ceiling-level amount of accuracy at training length on the tasks tested (hierarchy adds +0.012 on parity at $L = 16$, MDE 0.002, 16/16, from 0.988 to 1.000), and recursion does so where the base model trains unreliably. On Match-Query, where seed-paired contrasts are unreliable, a four-pass loop on a path-integrated model

Table 27: Loop and path integration on Match-Query 128² (explore with observations, then predict observations from actions alone), query length 256, $n = 8$ per arm (seeds 0–2 of all arms from one launch; seeds 3–7 of the two one-layer path-integrated arms from a second, and of the other three arms from a third), chance 0.0625. Index = token-index position; path-integrated = MapFormer-WM; loop = one shared layer applied four times. Interaction = loop gain with path integration minus with index. MDE = minimum detectable effect, $2.8 \text{sd}/\sqrt{n}$; unmeasured = within it. The paired contrasts are paired by seed as originally analysed; same-seed retrains drift by 0.185 on this task, so the paired MDEs are not reliable (see text). Unpaired: se = standard error, $t = t$ statistic. Reading: the loop’s gain survives unpaired; the interaction is not re-established unpaired.

arm	parameters	mean	sd	worst seed
index, one layer	204,182	0.108	0.025	0.06
index, loop $\times 4$	204,182	0.207	0.032	0.15
path-integrated, one layer	204,630	0.456	0.220	0.11
path-integrated, loop $\times 4$	204,630	0.870	0.099	0.77
path-integrated, 3 layers	601,174	0.771	0.263	0.14
paired: loop on path integration +0.414 (MDE 0.277, 7/8); loop on index +0.099 (MDE 0.045, 8/8)				
paired: interaction +0.315 (MDE 0.281); loop minus 3 layers +0.099 (MDE 0.252, 5/8), unmeasured				
unpaired, looped arm pooled over two batches ($n = 16$, 0.803 ± 0.200): loop +0.346 (se 0.092, t 3.75)				

survives an unpaired analysis (+0.346, se 0.092, t 3.75, looped arm pooled over two batches, $n = 16$ against 8); the paired estimates are +0.414 on the path-integrated model and +0.099 on an index model, with a paired interaction of +0.315 (MDE 0.281) that has not been re-checked unpaired. With $r = 4$ the looped model scores 0.986 ± 0.020 in its batch. No loss-matched analysis exists on Match-Query, so that gain is associated with, not separated from, more reliable training. On the torus the loop’s training-length gain is +0.052 raw but +0.006 at matched loss. An explicit what/where gate separates action from observation tokens by a factor of 4.16 and changes torus accuracy at $T = 512$ by +0.004 (MDE 0.008). State correction is reported in Section 10.

9.1 Recursion

Where there is headroom. On the torus a one-layer path-integrated model is already near ceiling, so a loop has nothing measurable to add there. Match-Query at 128² leaves headroom. Five arms, eight seeds each, pooled from three launches of the same recipe (Table 27).

A retrain of the looped arm with the same seeds did not reproduce it (mean per-seed drift 0.185), so these contrasts are read unpaired. The loop effect survives: pooled over the two batches the looped arm scores 0.803 ± 0.200 with one failure in 16 draws, and the loop on the path-integrated model minus no loop is +0.346 (se 0.092, t 3.75). The super-additive interaction (+0.315) is a paired estimate and has not been re-established unpaired. Unpaired, the loop against three real layers is +0.032 (se 0.105, t 0.30): not distinguishable from them, at a third of the parameters. An earlier reading that the looped arm never fails (every seed at least 0.77) came from one batch and is retracted. With the rank repair (Section 7.2), $r = 4$ plus the loop scores 0.986 ± 0.020 (worst seed 0.941) at 204,757 parameters, the best arm in its batch; rank adds +0.154 (MDE 0.084, 8/8) with the loop and +0.005 (MDE 0.091) without it, a paired interaction of +0.149 (MDE 0.112), subject to the same pairing caveat.

On parity the composition is additive. Sixteen seeds per arm, four depth settings in each position code: the loop minus one layer is +0.056 (MDE 0.014, 16/16) for the index code and +0.070 (MDE 0.056, 14/16) for path integration at $L = 128$; the difference, +0.014, is not detectable. The super-additive (paired) estimate is specific to Match-Query. For the index code the loop also beats three real layers (+0.010, MDE 0.003, 16/16).

On the torus the gain is convergence. In a twelve-seed batch on the torus, the looped model minus **Vanilla** at $T = 128$ is +0.052 raw (MDE 0.048, 12/12) but +0.006 loss-matched (MDE 0.017), with $r(\text{loss}, \text{accuracy}) = -0.956$ at that length. The looped model converges more reliably (mean final loss 0.0076 against 0.1549), and in a separate recipe study it converged on 8/8 seeds in every condition while **Vanilla** converged on 3/8, 4/8 and 6/8. The loop’s damage beyond training length (Table 29) is a separate matter (Appendix I).

Prior art. That recursion substitutes for depth is the premise of Mixture-of-Recursions (Bae et al., 2025). What this report adds is the interaction with path integration on one task, and the decomposition of the torus gain into convergence. A per-position oracle router on Match-Query would add at most +0.007 against a seed sd of 0.152 (Appendix I).

9.2 Gates and alternative generators

An explicit what/where gate. If the research goal is a where kept separate from the what, the obvious engineering move is to build the separation in. **Gated_r4** multiplies MapFormer’s signed increment by a per-token sigmoid (CoPE’s selection applied to MapFormer’s direction, 258 parameters; eight seeds per task, one batch per task). The gate separates: it passes 0.987 ± 0.003 on actions and 0.244 ± 0.043 on observations, a ratio of 4.16 (per seed 3.33–5.44, 8/8), against a pre-set floor of 1.35 taken from Selective RoPE’s gate and 1.00 for a frozen twin. It does not help. On the torus at $T = 512$, **Gated_r4** minus **Vanilla_r4** is +0.004 (MDE 0.008, 5/8), a powered negative against any gain the size of rank’s +0.038 there; at $T = 1024$ it is +0.003 (MDE 0.024), unmeasured. On recency at $T = 2048$ it is -0.039 raw (MDE 0.079) and -0.016 loss-matched (MDE 0.055), unmeasured. The only detectable contrast, the gate over its frozen twin (+0.009, MDE 0.008), is the size of the frozen twin’s own deficit. The pre-registered falsifier (no detectable effect at any length on either task) fired. This suggests that MapFormer’s bottleneck already separates the two token types: in ungated MapWM the action increment is 25.0 ± 22.4 ($r = 2$) and 35.9 ± 19.0 ($r = 4$) times the observation increment (Table 6, a separate batch).

Selective RoPE’s generator. Implemented in MapFormer’s placement (not a faithful Selective RoPE), the full generator is not better than MapFormer’s on parity (-0.009 against **Vanilla** at $L = 128$, MDE 0.017, unmeasured) and is marginally better on the torus at $T = 512$ (+0.031, MDE 0.030, 7/8, just outside its MDE), unmeasured at $T = 1024$ (+0.048, t 1.36), with more parameters (16,385 more on parity). Its per-knob arms cannot be attributed to single knobs, because each also deletes MapFormer’s frequency readout (Appendix F).

A forget gate. A FoX-style learned decay bias on MapWM at $r = 2$ adds +0.081 at $T = 1024$ (MDE 0.080, 7/8), borderline detectable, and nothing at $r = 4$ (-0.002). A frozen-bias control lands on **Vanilla** (-0.016 , MDE 0.118), and the direct live-minus-frozen contrast is +0.097 (MDE 0.105), unmeasured. The gain is anti-correlated with the learned decay ($r = -0.516$; five of eight seeds learn a negative decay), so it does not work by forgetting, and a transient-training-aid account was refuted against its registered sign ($r(\text{peak decay, gain}) = -0.531$). The mechanism is unidentified; a registered test of a clock account must be re-run (Section 14).

9.3 Hierarchy

Compositional transfer. On a motif task built so that a pooled summary is a sufficient statistic (Appendix D), a parameter-identical hourglass minus its flat scaffold is +0.136 (MDE 0.173, 7/8) at $T = 256$ and +0.151 (MDE 0.155, 7/8) at $T = 512$ under the better recipe: directional at $n = 8$, unmeasured. On the same task the recipe change alone is +0.160 (MDE 0.140, 7/8), larger than hierarchy’s effect.

Parity at length. Trained at $L = 512$ with twelve seeds, hierarchy minus flat is +0.060 (MDE 0.075, 12/12) unshared and +0.071 (MDE 0.081, 11/12) weight-shared. Both are inside their MDE and consistent in sign on nearly every seed; by this report’s rule they are directional. At $L = 1024$ and 2048 the effect is not distinguishable. At $L = 16$, where pooling leaves eight coarse tokens, hierarchy adds only a ceiling-level +0.012 at the training length and nothing measurable beyond it (Appendix D).

On MiniGrid it compensates. With commanded actions, hierarchy’s gain at $T = 1024$ is largest for a weak arm (+0.096, 8/8, RoPE with index position, base 0.827) and smallest for the strongest (+0.002, 5/8, PoPE with index position).

Table 28: Explicit state correction under drift, Match-Query with stochastic transitions (in the explore phase the executed action is resampled with probability p , so the recorded path drifts from the true one): **Level15** (MapFormer-WM plus a wrapped Kalman filter on the path integral) minus **Vanilla** (MapFormer-WM), paired by seed, as delta (MDE, seeds with a positive difference); MDE = minimum detectable effect, $2.8 \text{sd}/\sqrt{n}$; change with drift = $p = 0.10$ minus $p = 0$. $n = 8$ per arm, one batch per take. Take 1: 300 epochs, lr 3×10^{-4} . Take 2: 600 epochs, lr 10^{-3} . Arms are not converged at $p = 0.10$. Non-positional floor at $p = 0.10$ about 0.15. Reading: the prediction that the filter’s benefit grows with drift was not supported in either take; the slope itself is unmeasured.

take	$p = 0$	$p = 0.10$	change with drift
1	+0.035 (MDE 0.264, 5/8)	+0.038 (MDE 0.068, 5/8)	+0.003
2	+0.146 (MDE 0.326, 5/8)	+0.005 (MDE 0.065, 5/8)	−0.141

Compute. Hierarchy is an efficiency property. At $L = 2048$ the pooled scaffold takes 22.9% less time and 19.9% less memory than its flat twin; with weight sharing added, the combination saves 22.8% time and 19.9% memory with 66.5% fewer parameters. At $L = 16$ hierarchy is 12.2% slower. On byte-level text, a single-seed parameter-identical pair gives the hierarchical model 0.0032 bits per byte worse at $1.23\times$ the throughput (Appendix J).

10 Explicit state correction: no measurable benefit

The project’s original extension of MapFormer was explicit correction of the path integral: a wrapped invariant extended Kalman filter on $SO(2)$ (**Level15**: a learned constant prior variance and a per-token measurement noise). Its premise is drift in the integrated position together with observations that carry information about the true position. On the paper’s task neither is present. So stochastic transitions were added to Match-Query’s explore phase: the recorded action is the commanded one, the executed action is resampled, and observations reflect the true cell. At $p = 0.10$ the recorded path drifts by 13.05 cells from the true one. The n -gram gates hold at their clean levels (orders 1/3/5: 0.0647/0.0628/0.0648 against 0.0640/0.0620/0.0731 clean, chance 0.0625); the never-moved predictor reaches 0.1042, above chance, with no corrected clean 128^2 value to compare it with. The task has headroom.

The prediction and the result. The pre-registered prediction was that **Level15** minus **Vanilla** grows with drift. It was not supported in either of two recipes (Table 28). At $p = 0.10$ the filter’s benefit is +0.038 (MDE 0.068) in the first take and +0.005 (MDE 0.065) in the second: a benefit larger than about 0.07 at $p = 0.10$ would likely have been detected, in arms that are not converged. The change from $p = 0$ to $p = 0.10$ is +0.003 and −0.141. The $p = 0$ contrasts have MDE above 0.26, so the slope itself is unmeasured.

Two caveats travel with this result. The arms are not converged at $p = 0.10$ even at the larger budget. And destroying the query actions leaves about 45% of the $p = 0.10$ score in place (a local observation prior), so the non-positional floor there is about 0.15, not chance; both filter contrasts are differences of two numbers near that floor. The loop, by contrast, is detectable at $p = 0.10$ in both takes: +0.121 (MDE 0.065, 8/8) and +0.057 (MDE 0.025, 8/8).

What the filter does do. On the clean torus at $n = 5$, **Level15** minus **Vanilla** is detectable only beyond training length and, at this n , only after loss-matching: +0.062 (t 3.08) at $T = 512$ and +0.124 (t 3.83) at $T = 1024$; raw, both are inside their MDE (+0.057, MDE 0.150; +0.120, MDE 0.180). At $n = 12$ in a later batch the same contrast at $T = 1024$ is +0.129 raw (MDE 0.103, 10/12), detectable, and +0.083 loss-matched (MDE 0.083, 9/12), at its MDE; “only after loss-matching” is an $n = 5$ pattern. No named component is load-bearing: removing the measurement head, the per-token gain or the learned prior one at a time gives no detectable change (for example, full filter minus the wrap alone is +0.073, MDE 0.105, at $T = 1024$), and fixing the prior by the Riccati equation instead of learning it makes no detectable difference. Adding the filter inside a loop leaves accuracy below the filter alone beyond training length (0.830 against 0.878 at $T = 1024$). What survives is a stabilisation effect confined to out-of-distribution length, with no evidence that it works by inference from observations.

Table 29: Filter and loop on the clean torus: accuracy predicting the observation at revisited cells of a held-out map, trained at $T = 128$; mean over $n = 12$ per arm, one batch, 300 epochs. **Vanilla** = MapFormer-WM; **Level15** = plus a wrapped Kalman filter; **Looped** = one layer applied four times; **Level15Looped** = both; **LoopedSampled** = pass count drawn per batch. Blank floor 0.506. Rule 9, $r(\text{loss, accuracy})$: $r = -0.956, -0.471, -0.326$ at the three lengths. Reading: beyond training length the filter alone is best, and the filter inside a loop is below it.

arm	$T = 128$	$T = 512$	$T = 1024$	mean final loss
Vanilla	0.947	0.876	0.749	0.1549
Level15	0.990	0.953	0.878	0.0420
Looped	0.999	0.872	0.730	0.0076
Level15Looped	0.994	0.929	0.830	0.0189
LoopedSampled	0.997	0.905	0.745	0.0180

Scope. The claim is scoped to the tasks tested. On a landmark torus, where correction was first reported to help, a filter-free extra attention head ties the filter (+0.010, t 0.79, $n = 3$, exploratory; Appendix H), which withdraws the Kalman reading of that gain.

11 The length axis: an open problem

Nearly every effect in this report is concentrated beyond the training length. Rank adds +0.038 at $T = 512$ and +0.085 at $T = 1024$; the sign’s accuracy effect is unmeasured at $T = 128$ and detectable at $T = 512$ and 1024; the filter helps only beyond training length; the forget gate adds +0.022 raw at $T = 512$ and +0.081 at $T = 1024$; PoPE’s magnitude on top of path integration adds +0.007, +0.037 and +0.101 at $T = 128, 512$ and 1024 (grid 64, raw; the last two detectable, MDE 0.032 and 0.065); and MapEM degrades more slowly than MapWM with length on the paper task (Section 8.2; exploratory, convergence gate failed). No account covers these.

What the accumulator exponent covers. The growth exponent α tracks sign and rank (Section 7.4). It does not detectably change under the forget gate (−0.051, MDE 0.092), PoPE (+0.006, MDE 0.177) or the filter (+0.009, MDE 0.191), all unmeasured. A natural guess, that the filter’s wrap bounds the accumulator, is false: the filter wraps the innovation, not the state, and the ranges of the corrected and uncorrected phase are 285.6 and 283.9 at $T = 1024$.

A refuted import. The long-context account in which low-frequency channels never complete a cycle during training and are read at unseen phases later predicts that ablating those channels costs less out of distribution. Ablating the 32 lowest-frequency of the 64 channels (2 heads $\times$ 32) costs −0.001 at $T = 128$ and -0.139 ± 0.092 at $T = 1024$ for the signed $r = 4$ arm, -0.170 ± 0.084 for **Vanilla_r4** and -0.254 ± 0.164 at $r = 2$: more, not less.

PoPE’s gain follows length, not octaves. MapPoPE’s gain over MapWM was predicted to rise with the number of frequency octaves the map requires (grid size). Loss-matched at $T = 1024$ it is +0.077, +0.095 and +0.079 at grids 32, 64 and 128: flat. At fixed grid it rises with length in all three grids.

Collisions. In one arm (single-origin MapEM on the paper task, four seeds, four lengths), a failure at length is associated with the kernel ranking some distractor key at least as high as the correct key. The collision rate rises from 0.058 at $l = 256$ to 0.174 at $l = 2048$; accuracy is 0.973 with no collision and 0.787 with any. The relation is a switch, not a dose (one or two collisions: 0.743; 129 or more: 0.797), and collision rate accounts for 53% of the accuracy drop from $l = 256$ to $l = 2048$. This is an association within one arm; a cross-arm test has not been run.

12 Nested hierarchies: Dyck-2, and what its metric measures

MapFormer v4 Sec. 5.3 reports that single-layer MapFormers generalise perfectly on Dyck-2 while baselines need a second layer and fail out of distribution. We rebuilt the task to its appendix (two bracket types, a fixed length L whose maximum depth is exactly D , returning to depth 0; the F1 valid-continuation metric of Goodale et al.), trained at $L = 32$, $D = 4$ on 560k sequences at their recipe, and evaluated on their $L \times D$ grid. Eight seeds per arm, one batch. Pre-registration: `DYCK_PREREG.md`.

Table 30: Dyck-2 F1 (ours, 8 seeds; paper value in brackets). Every model fits the training cell; the split appears only out of distribution. The n -gram floor is the best of orders 1–6 fitted on the training distribution: a lookup table with no stack. The paper reports no such floor.

arm	$L32\ D4$ (trained)	$L128\ D4$	$L32\ D12$	$L128\ D12$
MapPoPE-1L	0.988	0.923	0.976	0.927
MapWM-1L	0.985 [1.00]	0.900 [0.97]	0.942 [0.98]	0.868 [0.94]
MapEM-1L	0.985 [1.00]	0.878 [0.99]	0.949 [0.98]	0.888 [0.95]
CoPE-1L	0.900	0.703	0.854	0.703
PoPE-1L	0.911	0.647	0.882	0.615
RoPE-1L	0.910 [0.91]	0.575 [0.78]	0.822 [0.86]	0.566 [0.77]
RoPE-2L	0.947 [0.97]	0.521 [0.59]	0.691 [0.66]	0.498 [0.50]
n -gram floor (no stack)	<i>0.904</i>	<i>0.896</i>	<i>0.903</i>	<i>0.884</i>

The ordering replicates; the levels do not. MapFormer minus RoPE-2L at the hardest cell is +0.370 (MapWM) and +0.390 (MapEM), 8/8 seeds, against the paper’s +0.44. RoPE-2L reproduces closely (8 of its 16 cells within 0.03). But every MapFormer cell is below the published value (mean -0.045 for MapWM, -0.053 for MapEM), so we reproduce the direction and not the magnitude. Batch size is the paper’s main unstated hyperparameter and remains the untested suspect.

The metric gives most of its range to positions that need no stack. Both (and [are legal at every position, so a predictor collects most of the score without reasoning; and where a closer *is* legal, the previous token is the top of the stack two thirds of the time. A one-token lookup therefore scores 0.857 at $L128\ D12$ — above every index model — while being at chance for choosing the legal closer beyond distance 2. Scoring the same predictions with the metrics the Dyck literature uses separates the models properly (Table 31).

Table 31: The same checkpoints under the standard Dyck metrics. *Bracket-closing memory* (Hewitt et al. 2020; Yao et al. 2021 follow it): probability of the legal closer renormalised over closing brackets, averaged over distances; chance 0.500. *Valid-set prediction per sequence* (Suzgun et al. 2019; Bhattamishra et al. 2020; Ebrahimi et al. 2020): correct only if every step of the sequence is correct; chance 0.000.

arm	bracket-closing memory		set prediction, per sequence	
	$L32\ D4$	$L128\ D12$	$L32\ D4$	$L128\ D12$
MapPoPE-1L	0.994	0.719	0.661	0.042
MapWM-1L	0.992	0.638	0.672	0.003
MapEM-1L	0.980	0.620	0.615	0.000
PoPE-2L	0.919	0.578	0.084	0.000
RoPE-2L	0.914	0.574	0.097	0.000
PoPE-1L	0.646	0.551	0.000	0.000
RoPE-1L	0.627	0.535	0.000	0.000
n -gram (no stack)	<i>0.531</i>	<i>0.508</i>	<i>0.020</i>	<i>0.000</i>

Three things follow. First, the paper’s real claim survives and sharpens: a *one-layer* path-integrated model learns the stack at the training distribution (0.98–0.99) where one-layer index models do not (0.63–0.65), and index models need a second layer to learn it, exactly as the transformer theory predicts — a distinction the F1 metric hides by scoring RoPE-1L at 0.912 (the 0.910 of Table 30 is the same arm on the larger 1024-sequence evaluation). Second, “perfect OOD generalisation” survives no standard metric: under the

published per-sequence criterion MapWM falls from 0.672 to 0.003. Third, only under the paper’s own F1 does the stack-free n-gram outscore every index model.

The floor verdict, stated. On the paper’s own metric at the hardest cell, neither headline MapFormer beats the stack-free n-gram: MapWM is -0.016 under it and MapEM $+0.003$ over it, and MapPoPE’s $+0.042$ has MDE 0.046, i.e. formally at the floor. Only the standard metrics of Table 31 separate the models from a lookup table.

Reach, not capability. Splitting closer accuracy by how far back the bracket to be closed was opened shows each model has a horizon rather than an ability: at $L = 128$, $D = 12$ all arms are near-perfect within two tokens, and at distance 33+ MapPoPE is at 0.730, MapWM 0.611, RoPE 0.539 and the n-gram 0.511. Pushed to $L = 512$ every model reaches chance beyond roughly 128 tokens of reach (DYCK_FAR_PROBE.md; that evaluation was run but uncommitted when this section was first written, and is committed now).

13 PoPE’s own tasks, and why the two mechanisms anti-compose

PoPE (Gopalakrishnan et al., arXiv:2509.10534) decouples what from where by putting content in the *magnitude* and position in the *phase*. Its phase offset δ_c is a per-head constant and its amplitudes are non-negative. MapFormer instead rotates the content vectors, so the score’s phase carries $\angle q_p - \angle k_p$: the current query and key can shift the kernel’s peak, and the amplitude can cancel. We replicated two of PoPE’s results and crossed them with path integration.

Indirect Indexing replicates, and the task is a late-transition search problem. At the paper’s 100k-iteration budget our PoPE solved 1/8 seeds against their implicit 3/3. Doubling the budget takes PoPE to 7/8 (solver mean 0.965 against their 0.948 ± 0.029) and MapPoPE to 8/8, so the paper’s result stands and our shortfall was the budget. A run either transitions (lift-off between 40k and 190k steps) or sits at 0.09 forever. Path integration’s advantage at 100k (5/8 versus 1/8) is therefore *speed*: at an adequate budget the two tie (Fisher $p = 1.0$), and the ~ 15 k-step lift-off difference is unmeasured at $n = 7$. Out of distribution neither encoding generalises past the trained shift range (chance at $|k| \in [16, 30]$).

Bach Chorales replicates, and path integration adds nothing in distribution — under a ceiling worth three times the effect. Pitch transposition (the augmentation PoPE’s paper applies to MAESTRO and not to JSB) gains 0.107 NLL for PoPE and 0.090 for MapPoPE, 5/5 seeds, against the 0.032 separating PoPE from RoPE; the ordering is unchanged and the PoPE-over-MapPoPE gap widens to 0.028.

The replication itself. PoPE beats RoPE by -0.032 NLL (5/5 seeds; paper -0.019), with levels within 0.025 of published. Path integration is unmeasured on both rows (-0.005 on the RoPE row at 3/5 seeds, $+0.011$ on the PoPE row at 0/5). This is the first clean null for path integration on a natural-sequence task, and it was registered with no predicted direction.

Beyond the training context the ordering inverts, and the naive combination collapses. Training on 512-token crops and scoring whole pieces by position bucket (Table 32), path integration wins and the win grows with distance (-0.662 at $2-4\times$, 5/5 seeds), PoPE’s decoupling is itself a length win on the index row (-0.461), and *MapPoPE—best of all arms in distribution in these 512-crop runs, though at the MDE boundary, and worse than PoPE on 5/5 seeds at full context—is the worst by far outside it.*

13.1 Why the combination fails, and what fixes it

The accumulator is identical in both arms. On music the increments do not cancel, so $S_t = \sum \Delta$ is a *clock*: the growth exponent is $\alpha = 1.003$ for MapPoPE and 1.005 for MapWM, and the range quadruples at $4\times$ length in both. The obvious story — MapPoPE’s angle runs away — is therefore ruled out.

Table 32: Bach Chorales trained on 512-token crops; test NLL by position bucket, lower is better, 5 seeds. The last row adds a per-token phase to MapPoPE (Section 13.1).

arm	position $\times$ encoding	0–512 (trained)	512–1024	1024–2048
RoPE	index, RoPE	0.5512	0.8116	2.0587
PoPE	index, PoPE	0.5403	0.6749	1.5973
MapWM	path int., RoPE	0.5382	0.7835	1.3969
MapPoPE	path int., PoPE	0.5235	1.9536	4.6158
MapPoPE + per-token phase	path int., PoPE ⁺	0.5182	0.5629	0.6162
MapPoPE + decay envelope	path int., PoPE	0.5498	0.5860	0.6223
PoPE + decay envelope	index, PoPE	0.5580	0.5965	0.6262

A dissociation locates the difference. Compressing ω at evaluation, the path-integration analogue of position interpolation, *improves* MapPoPE out of distribution by 1.70 nats and *worsens* MapWM by 1.44, and at scale 0.25 the two are equal. Same accumulator, opposite response: MapWM’s robustness is not that its argument stays in range but that content carries a second degree of freedom which compression destroys.

Two reparameterisations fail to explain it. Rank: the collapse survives $r = 1, 2$ and 4 (4.08, 4.62, 4.63) and the rank effect is equally present on the MapWM control row — where its direction is $r1 < r2 < r4$ (0.912, 1.397, 1.660 at 2–4 $\times$), inverting this project’s torus-derived “use $r = 4$ ” recommendation and recorded as an unresolved task-dependent conflict. The ω base: the predicted direction inverts — base 512 is best and 32768 worst on *both* rows — so raising the base, the standard fix for index RoPE, is not the fix here. MapPoPE never beats MapWM out of distribution at any rank or base (0/5 seeds in every cell).

Two interventions do. Centring the increment, which shrinks the accumulator’s excursion ($\alpha 1.00 \rightarrow 0.89$, range $3\times$ smaller — not the random walk it was designed to be), takes MapPoPE from 4.616 to 0.911 and MapWM from 1.397 to 0.741 — a gain $5.7\times$ larger for the arm that cannot compensate, which was the registered discriminator. Giving δ a per-token component, leaving the accumulator untouched, takes MapPoPE to 0.733 with a parameter-matched inert twin flat (−0.167, MDE 0.828); forcing that phase away from zero at initialisation improves it further to 0.616, better than MapWM at every bucket.

A decay envelope repairs it for 48 parameters, and its damage has two separable causes. An ALiBi-style envelope (`scores` $= \text{softplus}(\lambda_h) \cdot d$, one scalar per head) removes the collapse as well as the per-token phase does: 0.6223 against the phase’s 0.6162 at 2–4 $\times$, for 48 parameters rather than 786k. It repairs the index row too (1.597 $\rightarrow$ 0.626), so much of the pathology is the generic long-context problem rather than something specific to a learned accumulator. But the envelope needs a notion of distance, and on Dyck-2 the choice is decisive: decaying over token distance drives an index model’s long-range bracket matching to chance (0.503), decaying over the model’s own learned state does not (0.784), a gap of +0.281 on 8/8 seeds. Two controls split that gap. Weakening the token envelope to the same effective penalty (26 logits at distance 64, against 2) recovers +0.136 (8/8), so roughly half the damage is simply over-applied locality; at matched strength the learned-state metric is still +0.145 ahead (8/8), so the metric is not reducible to steepness. A *frozen* state metric is indistinguishable from weakening the envelope. Caveat: final training loss orders with accuracy across these arms ($r = -0.995$) and the losses do not overlap, so the residual cannot be loss-matched.

The boundary does not survive a within-task test (2026-09-20). Forcing Dyck’s accumulator from a map into a clock — monotone increments, $\alpha 0.6 \rightarrow 1.05$ verified, same task, same recipe — reproduces neither half of the account: PoPE’s encoding helps *more* on the clock (+0.159 against +0.058; MDE 0.244, so the registered negative simply fails to appear), and the per-token phase hurts *more* (−0.143 against −0.046, 0/8 seeds, detectable), where the registered predictions were the opposite in both cases. The three Bach interventions below stand as measured; the generalisation from them to a clock/map rule across tasks is withdrawn, and the excursion’s absolute size (551 against 24), model size, frequency count and local solvability are all uncontrolled alternatives.

The between-task pattern, which prompted that test. The same per-token phase pays only where the accumulator leaves its trained range. On bounded accumulators it is neutral to harmful: Dyck-2 -0.046 against its own inert twin (MDE 0.043 ; the forced-initialisation arm gives -0.047 but is unmeasured), and the torus 0.911 against 0.963 at $l = 2048$ with a seed spread eight times the baseline’s — a ceiling condition, where only the forced-initialisation cell is detectable. Forcing the phase harder helps on music and hurts on the torus; on Dyck the two initialisations are indistinguishable.

What is withdrawn. Three of our own accounts died here and are recorded rather than removed: that the collapse is a rank or frequency-magnitude effect (refuted by the two sweeps above); that restoring a content-dependent phase must cost pure-indexing accuracy by re-entangling what and where (Indirect Indexing is $5/8$ with and without it); and that a model declines the freedom where it is useless (on Dyck it keeps 0.825 rad, more than the music model’s 0.412 , and is worse for it). What survives is narrower: the collapse needs both an out-of-range accumulator and a kernel that cannot compensate, and each half now has an intervention behind it.

14 Limitations

- **Transfer across a change of structure is not measured.** Every evaluation redraws the observation map within one topology and action space. The TEM claim that motivates the work is about carrying structural knowledge across structures; nothing here tests it, and sample efficiency (learning curves) is not measured either.
- **No language result.** The language experiments are underpowered or single-seed (Appendix J), and published work reports negatives for content-dependent phase on language (HGRN, HGRN2) alongside MapFormer v4’s own mixed result.
- **One varying-offset task.** Claim 4 rests on k -back recency. MapWM at the larger budget was not run.
- **Out-of-distribution length.** Most accuracy effects are concentrated beyond the training length (Section 11), and no account of that concentration exists.
- **Recipe dependence of index baselines.** Under the paper’s recipe the torus index arms are undertrained. Under the converged recipe the raw position main effect at training length is $+0.243$ (MDE 0.038), against $+0.461$ in the paper-recipe batch (different evaluator); it stays detectable. Loss-matched readings depend on the pool, and they are uninformative where arm losses do not overlap (Section 5.4).
- **Conditions established on different tasks.** The three conditions of the thesis were shown on different tasks; no single task shows them binding together, and they are not claimed to be sufficient.
- **Boundaries at $n = 3$.** The MiniWorld map-extent threshold, the 12-heading budget curve, the family-tree non-commutativity result and several appendix results are exploratory.
- **Holdability at a pinned point.** The held rewind was shown with the content columns of W_{in} fixed at zero.
- **Simulator realism.** Allocentric recoding needs a known heading, and the one realistic simulator measured (Habitat) moves the agent by a different amount than commanded on 69–91% of forward steps.
- **Match-Query reproducibility.** Match-Query does not reproduce across batches or same-seed retrains, so its seed-paired contrasts are unreliable; the loop effect is cited unpaired and the paired interactions are not re-established.
- **Borrowed benchmarks do not discriminate this axis.** On Flip-Flop (as defined by Golovneva et al. (2024)) every contrast is unmeasured, which the per-offset recency curve predicts: Flip-Flop asks only for the most recent write, where index codes score 0.99 . MQAR (Arora et al., 2023) did not train at this scale for any arm (Appendix J).

- **A lost batch.** A completed, pre-registered test of the forget gate as a second, monotone accumulator was deleted before analysis and must be re-run; the forget gate’s mechanism remains unidentified.

15 Conclusion

On tasks that redraw the observation map, MapFormer’s transferable where is its accumulated, content-driven phase. It is the larger factor on raw accuracy on MapFormer’s own task, at training length under the paper’s recipe and at all three lengths under a converged one; the rotary encoding adds a smaller amount beyond the training length. The phase pays when the input makes each token’s displacement fixed, when increments are signed and well conditioned so that they cancel, and when training finds the solution. The first two conditions were supported by single-operation interventions (the action record, the absolute value, the rank), though that rank acts through conditioning is descriptive; the third by constructing the solution, holding it, and measuring where from-scratch training finds it. A fully factorised where, MapEM’s shared kernel, represents the one varying-offset task measured exactly but is found per query token, and giving the kernel per-pair freedom changes both the accuracy and the mechanism. Filtering and gating the where add no measurable accuracy at training length on these tasks, and pooling’s only detectable gain is ceiling-level (its larger gains are directional); recursion does, in association with more reliable optimisation (on the torus its gain vanishes at matched loss; Match-Query was not loss-matched). What remains open is why so many mechanisms help only beyond the training length, and whether any of this transfers across a change of structure.

A Gates and floors

Gate records. Table 33 collects the shortcut gates of every task used in the report. Each was run on the task code before training, except where stated.

Table 33: Shortcut gates and floors, run on the task code before training unless stated: scores of trivial predictors on the scored events, which must sit at chance or the floor. o1/o2/o3/o5 = n -gram predictors of those orders on the action or answer stream alone; marginal = always the most common answer; never-moved, most-recent, echo, repeat-previous and hub = task-specific trivial predictors (the hub floor is a hub-node baseline); oracle = a check that the task is solvable; label mass = share or number of scored events. MiniWorld: G2 = marginal, G4 = best non-blank action n -gram, G5 = label mass, G6 = revisit lag. Context destruction pass = trained models collapse when the required context is destroyed. Reading: every task used passes its shortcut gates (copy has no dynamic range; separation breaks at $n_{\text{obs}} = 4$).

task	gate	verdict
Match-Query $n_{\text{obs}} = 16$	64^2 , action-stream n -grams o1/o3/o5 0.0625/0.0689/0.0664 against chance 0.0625; marginal 0.0677; never-moved floor 0.0893 (corrected from 0.0516, a scorer bug)	pass; context destruction pass
Match-Query 128 ²	n -grams 0.0640/0.0620/0.0731; never-moved 0.0490 (pre-fix scorer, not re-measured)	pass
Match-Query $n_{\text{obs}} = 4$	64^2 , n -grams 0.2479/0.2484/0.2428 against chance 0.25; never-moved 0.1515 (pre-fix scorer, not re-measured)	pass; separation breaks
Match-Query, $p = 0.10$ transitions	n -grams 0.0647/0.0628/0.0648; marginal 0.0685; never-moved 0.1042 (600 episodes)	pass; floor about 0.15
Parity	marginal, n -grams 1/2/3/5, echo, repeat-previous: worst excess +0.0122	pass
Copy	as parity	pass, but no dynamic range
Recency, $k_{\text{max}} = 64$	all rows pass at minimum gap 4/16/64 (800 episodes); most-recent floor 0.0771; answer 128.3 ± 11.8 tokens back at $k = 64$	pass
Recency, k from a small set	n -grams and oracle pass; most-recent 0.2958 ($m = 4$) and 0.1123 ($m = 16$)	primary excludes $k = 1$
Torus, rotate / allocentric	answer-stream o1/o2/o3/o5 0.501/0.472/0.440/0.462 against 0.507, with scoring on moves only	pass
D -dimensional torus	best action n -gram 0.513/0.531/0.536 against majority 0.506/0.523/0.526 ($D = 2, 3, 5$)	pass
Family tree, depth 5	hub floor 0.1628; marginal 0.1280; last observation 0.1240; n -grams 0.1235/0.1202/0.1241/0.1283; non-commutativity 1.000	pass; context destruction pass
Compositional motif	oracle 1.00; label mass 14.3/23.2/34.4% at $T = 128/256/512$; context destruction removes 76/80/67% of headroom	pass
MiniWorld, grid 32	G4 non-blank best 0.055/0.019/0.000 against marginals; G5 label mass 50.4 per trajectory and G6 median lag 33 identical across n_{obs}	pass (G2 warning at $n_{\text{obs}} = 256$)

Excluded task families. Two families of goal-directed demonstration tasks failed the action-stream gate and are not used. On a hierarchical-goal task, randomising the goal and the entire explore phase left accuracy unchanged (0.912 to 0.913 for **MapWM-Flat**), an action-only Markov predictor scored 0.969 at order 1 (and 0.971 at order 3 after an interleaving fix), and closed-loop success was 0.013–0.037 against a random floor of 0.010. Four planner-demonstration tasks were solvable from the action stream alone at 0.969, 0.969, 0.791 and 0.650 against chance 0.250. Validating a fix at order 1 only had certified a stronger order-3 shortcut; gates are therefore run at orders 1–5.

Why index models exceed the paper-task floor. Table 34.

Table 34: Paper task, paper’s recipe: revisit accuracy (predicting the observation at a revisited cell) by steps since the last visit to the cell. **Vanilla** = MapFormer-WM; **RoPE**, **PlainFlat** = index position. Blank rate = per-bin floor; share = fraction of scored events. Seed count not recorded; exploratory. Reading: index models beat the blank rate only at interval 1–2, a retrace.

	1–2	3–4	5–8	9–16	17–32	33–64	65+
Vanilla	0.995	0.996	0.996	0.993	0.980	0.937	0.881
RoPE	0.557	0.516	0.505	0.503	0.500	0.495	0.507
PlainFlat	0.575	0.518	0.505	0.501	0.477	0.475	0.488
blank rate	0.504	0.514	0.514	0.509	0.484	0.504	0.533
share of events	19.3%	16.3%	24.7%	24.6%	8.9%	4.5%	1.7%

Match-Query at scale and at length. At 128^2 ($n = 3$, older recipe) path integration scores 0.823 ± 0.043 against **PlainFlat**’s 0.192 ± 0.022 with no seed overlap. With four observation types (64^2 , chance 0.25) the separation breaks: 0.510 ± 0.187 against 0.332 ± 0.012 , with one path-integrated seed (0.321) inside the index range. On three seeds of the base model, path integration degrades gracefully with blind query length (0.904 at 256 to 0.693 at 2048).

Noisy Match-Query context destruction. With $n = 4$ seeds per cell, clean Match-Query collapses under destruction (**Vanilla** 0.493 to 0.050 with explore observations destroyed and 0.086 with query actions destroyed). At $p = 0.10$, destroying query actions leaves 0.142 of **Vanilla**’s 0.302 and 0.167 of the looped model’s 0.366, so about 45% of the noisy score is non-positional.

The MiniWorld noise floor. A function-identical twin of **Vanilla** (a gate multiplied out, zero gate gradient) differs from **Vanilla** by a mean absolute 0.150 at $T = 512$ and 0.163 at $T = 1024$ (range -0.228 to $+0.410$, nine pairs). Only one of the nine triples was converged, so this floor overstates the noise of converged conditions.

B Early baselines and TEM

These results use the April 2026 linear-decay recipe at $n = 3$, without gates, and are exploratory. Seed-0 retrains under current code are bitwise identical to the stored checkpoints (a determinism check), which licenses the clean and action-noise configurations; landmark configurations of the same period are not used.

Table 35: April batch, torus paper task, fresh map: accuracy predicting the observation at revisited cells, trained at $T = 128$; mean $\pm$ sd, $n = 3$, 50 epochs linear decay, no gates; exploratory. “Noise” replaces 10% of actions during training (stochastic transitions), not at evaluation. Blank floor about 0.50. **Vanilla** = MapFormer-WM; **VanillaEM** = MapFormer-EM, separate-origin form; **RoPE** = MapFormer’s architecture with index position; **LSTM** = recurrent network; **MambaLike** = diagonal- A approximation of Mamba. Reading: exploratory early baselines, kept for reference.

arm	clean $T = 128$	clean $T = 512$	noise $T = 128$	noise $T = 512$
Vanilla	0.992 ± 0.006	0.913 ± 0.037	0.954 ± 0.008	0.739 ± 0.062
VanillaEM	1.000 ± 0.000	0.972 ± 0.003	0.957 ± 0.009	0.765 ± 0.138
RoPE	0.635 ± 0.072	0.463 ± 0.026	0.608 ± 0.070	0.469 ± 0.027
LSTM	0.860 ± 0.010	0.800 ± 0.004	0.798 ± 0.018	0.743 ± 0.010
CoPE (reimplemented)	0.741 ± 0.037	0.679 ± 0.024	0.687 ± 0.032	0.633 ± 0.022
MambaLike (diagonal A)	0.591 ± 0.005	0.573 ± 0.007	0.586 ± 0.003	0.568 ± 0.004

Length. On the clean configuration at $T = 2048$, evaluated on the training map rather than the fresh map of Table 35, **LSTM** (0.739) exceeds **Vanilla** (0.635), and **VanillaEM** scores 0.827; **MambaLike** stays near floor (0.558). The diagonal- A SSM is an approximation of Mamba (Gu and Dao, 2023); the paper’s block-diagonal MAMPa was not implemented.

TEM. A transformer formulation of TEM (Whittington et al., 2022), with two layer norms added to stabilise its recurrence, scores 0.946 ± 0.003 and 0.856 ± 0.006 at $T = 128$ and 512 on the clean task and 0.759 ± 0.011 and 0.668 ± 0.022 under action noise ($n = 3$). Those noise values apply the noise at evaluation as well, unlike Table 35; under that protocol **Vanilla** scores 0.757 ± 0.013 and 0.638 ± 0.035 , so the noise columns of Table 35 are not comparable with them. A faithful TEM with orthogonal action matrices and Hopfield memory scores 1.000 ± 0.000 and 0.966 ± 0.008 clean and 0.762 ± 0.003 and 0.709 ± 0.002 under evaluation-time noise ($n = 3$); adding a feed-forward network moves the clean $T = 512$ value to 0.969 ± 0.002 . The faithful-TEM values are scored on one fixed map (environment seed 0), not on a map drawn from a held-out seed. In the timing configuration it has 20,705 parameters against MapFormer’s 405,472 and scales $120.4\times$ in time over the length range of Table 7.

Stitching, ported from CSCG (exploratory). A scored stitching task built first was dropped before use: its negative control was defeatable without a map (0.617 balanced accuracy). In its place, models trained only on the revisit objective were probed; no shortcut gates were run on those training episodes. In two-room episodes where two cells emit the same symbol after a shared prefix, **MapWM-Flat** attends more to the correct look-alike cell than a paired confound by $+0.1306 \pm 0.0242$ (three seeds, per seed $+0.1033$, $+0.1495$, $+0.1388$), and **PlainFlat** by -0.0053 ± 0.0158 . The arms also differ in whether they learned the map (0.971–0.974 against 0.57–0.61) and in the observations along their approach, so this is not a clean architecture comparison and does not isolate path integration. CSCG (George et al., 2021) reports no number; this ports its evaluation style.

First paper-task pass. At $n = 3$ with the paper recipe, the separate-origin MapEM scores 0.898 ± 0.108 on the training map (per seed 0.778, 0.931, 0.986) and 0.901 ± 0.102 on a fresh map, against single-origin 0.987 ± 0.012 and MapWM 0.989 ± 0.010 , which are identical on both maps.

C Family tree and non-commutativity

The paper motivates a non-commutative MapEM (MapEM-NC: $K = n(n - 1)/2$ skew generators composed by a sequential product) with family trees, where “mother of” and “father of” do not commute, but does not run one. On a depth-5 ancestor tree (63 nodes, 8 observation types, 8 relational actions, train $T = 64$, OOD $T = 128$, hub floor 0.1628), five arms were trained in one batch at $n = 3$ (Table 36).

Table 36: Family tree, depth 5, $n = 3$ per arm, one batch, 2 layers, 100 epochs; held-out accuracy, trained at $T = 64$; hub floor 0.1628. **Level15** = MapFormer-WM plus filter; **MapWM-Flat** = MapFormer-WM; MapEM-NC-NL = non-commutative MapEM; **Plain-Flat** = index transformer. Contrasts at $T = 64$: delta (sd of paired differences, MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$, seeds with a positive difference); unmeasured = within the MDE. Exploratory. Reading: non-commutativity is real but small; path integration’s gain is larger.

arm	$T = 64$	$T = 128$
Level15	0.843 ± 0.015	0.789 ± 0.027
MapWM-Flat	0.805 ± 0.072	0.746 ± 0.080
MapEM-NC-NL (non-commutative)	0.729 ± 0.010	0.672 ± 0.012
single-origin MapEM (commutative)	0.715 ± 0.008	0.659 ± 0.015
Plain-Flat (index)	0.601 ± 0.011	0.550 ± 0.031
MapEM-NC-NL – single-origin MapEM: $+0.013$ (sd 0.005, MDE 0.008, 3/3)		
MapWM-Flat – Plain-Flat : $+0.205$ (sd 0.073, MDE 0.118, 3/3)		
MapWM-Flat – MapEM-NC-NL: $+0.077$ (MDE 0.111, 2/3), unmeasured		
Level15 – MapWM-Flat : $+0.038$ (MDE 0.111, 2/3), unmeasured		

Context destruction passes: shuffling observations drops MapEM-NC-NL from 0.7282 to 0.1626, the hub floor to three decimals. At depth 7 (255 nodes, $n = 3$), the non-commutative gains shrink to $+0.0017$ and $+0.0037$ (2/3 each) while single-origin (commutative) MapEM minus **Plain-Flat** grows from $+0.115$ to $+0.180$ (no MDE recorded). The group-theoretic argument for non-commutativity is correct; it does not translate into a performance gap here. In training time the linear variant MapEM-NC-L grows $14.4\times$ from $L = 128$ to $L = 2048$ against $3.9\times$ for single-origin MapEM; MapEM-NC-NL was not timed.

D Compositional task and hierarchy

Task. A 64×64 torus is tiled into 8×8 rooms, and four motif templates, redrawn every episode, are assigned to several rooms. *cross_nb_acc* scores cells whose motif was seen in another room but whose exact cell was not: a pooled summary of motifs is a sufficient statistic for it. Floor about 0.072; train $T = 256$; held-out environments.

Old recipe. Under the published recipe (50 epochs, linear decay, lr 3×10^{-4} , three layers), at $n = 8$: MapWM-Hier 0.415 ± 0.096 , parameter-matched flat MapWM-FlatHG 0.285 ± 0.067 , Plain-Hier 0.318 ± 0.029 , Plain-Flat 0.216 ± 0.004 . Arms accumulated over several batches, so no MDE is given. A variant whose coarse level uses index position scored 0.556 ± 0.123 (exploratory). MapEM with separate origins scored 0.097 ± 0.013 ($n = 3$).

Recipe versus architecture. Table 37. The published recipe reproduces (0.354 ± 0.070 against the published 0.415 ± 0.096 , same distribution); cosine decay at lr 10^{-3} for 150 epochs adds +0.160 (MDE 0.140, 7/8). Seed variance rose rather than fell, the opposite of the torus. A same-seed retrain of the hierarchy recheck reproduced arm C digit for digit, a determinism check.

Table 37: Compositional headroom, Hourglass_k2 (hierarchical MapFormer-WM), $n = 8$ per arm, one batch. *cross_nb_acc* = held-out accuracy on cells whose motif was seen elsewhere but the cell was not (floor about 0.072); final loss = final training loss. A–C vary only the recipe; D is a looped hourglass with 209,256 parameters against 605,800. Contrasts paired by seed; MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$. Reading: the recipe change alone is detectable and larger than any architectural effect measured on this task.

arm	cross_nb_acc at $T = 256$	final loss
A: linear, lr 3×10^{-4} , 50 ep	0.354 ± 0.070	0.7466
B: cosine, lr 10^{-3} , 50 ep	0.460 ± 0.151	0.5579
C: cosine, lr 10^{-3} , 150 ep	0.514 ± 0.126	0.4760
D: looped, cosine, lr 10^{-3} , 150 ep	0.542 ± 0.170	0.4095
C – A +0.160 (MDE 0.140, 7/8); B – A +0.107 (MDE 0.150); D – C +0.029 (MDE 0.166)		

Structure-aligned pooling does not help. Pooling on oracle room boundaries scored 0.254 ± 0.014 against the flat scaffold’s 0.281 ($n = 3$); adding a local frame reset at room entry dropped both transfer (0.157) and exact recall (0.94 to 0.77). Room-aligned pooling did not help ($n = 3$, exploratory).

Which lever pays depends on structure. Sweeping the number of motif templates (2/4/8/16, $n = 3$, old recipe), hierarchy’s transfer gain falls (+0.163, +0.135, +0.034, +0.006) while path integration’s exact-recall gain does not (+0.060, +0.158, +0.251, +0.154).

Parity at $L = 16$. Trained at $L = 16$, hierarchy minus flat at $L = 128$ is +0.001 (MDE 0.006) for the index code and -0.014 (MDE 0.109) for path integration ($n = 16$). With pooling by 2 leaving eight coarse tokens, the mechanism cannot operate at this length; the $L = 512$ result is in Section 9.3.

E MiniGrid and MiniWorld: supplementary

Results in this appendix at $n = 3$ are exploratory.

The other four torus knobs. At $n = 3$ (16 epochs, one layer), the position effect is +0.478 at baseline, +0.265 with an egocentric observation, +0.251 with walls, +0.324 on a small map and +0.299 with rich observations; all four combined with rotation give -0.084 , approximate because an order-3 answer-stream n -gram reaches 0.634 against a 0.536 marginal. The pre-registered prediction that aliasing and map size drive the effect was refuted, and two of these knobs carry one unstable seed each.

Frequency learning (exploratory). Path-integrated arms learn ω while index arms use fixed frequencies. In MapWM on MiniGrid, learned minus frozen ω is +0.004 ($T = 512$) and -0.008 ($T = 1024$) ($n = 3$, exploratory), so frequency learning does not measurably account for the position effect there.

Allocentric MiniGrid, full factorial at $n = 3$. With the allocentric record at $r = 2$, the eight-cell position main effect is +0.013 and +0.020 at $T = 512$ and 1024, against -0.005 and -0.021 for the commanded record (the latter from the $n = 8$ batch, so the comparison crosses batches). A path-integrated arm does not outrank every index arm (PoPE-Hier 0.817 against MapWM-Flat 0.809 at $T = 1024$). A reproduction control retrained PoPE-Flat to its previous values exactly. Allocentric absolute scores are lower than commanded ones (best 0.825 against 0.955 at $T = 1024$).

Twelve headings, per seed. Allocentric, $n = 3$ per budget: at 980 batches Vanilla scores 0.661/0.798/0.858 (final losses 1.357/0.881/0.629); at 2000, 0.885/0.893/0.894 (0.552/0.527/0.507); at 4000, 0.807/0.799/0.906 (0.834/0.815/0.422), where RoPE scores 0.542/0.555/0.555. With 0.15 rad actuation noise on each executed turn, measured only at the undertrained 980-batch budget, the effect falls from +0.263 to +0.230 (both from one evaluation; the budget-curve evaluation of the same 980-batch runs gives +0.264, the value in Section 6.2).

Habitat measurements. habitat-sim 0.3.3 (Savva et al., 2019): turns are exactly 30 degrees with zero displacement once configured (the library default is 10), forward is exactly 0.25 m, and 12 turns close the circle. Across three scenes (9.2, 52.9 and 226.7 m²), forward moves produce the exact displacement on 19.7%, 30.7% and 8.6% of attempts, slide partially on 51.5%, 48.5% and 78.7%, and are fully blocked on 28.8%, 20.8% and 12.7%. No model was trained.

MiniWorld with a fixed map. When the observation map is fixed per seed rather than redrawn, the position effect is below the 0.150 noise floor at both lengths (+0.020 $\pm$ 0.013 raw and +0.008 $\pm$ 0.009 allocentric at $T = 512$; $n = 3$, exploratory), and at $T = 512$ the allocentric record raises every arm (Vanilla 0.653 $\rightarrow$ 0.801, RoPE 0.620 $\rightarrow$ 0.798). A fixed map dilutes the need for path integration by design.

Environments. MiniGrid DoorKey-16x16 and MiniWorld OneRoom are from Chevalier-Boisvert et al. (2023).

F Rank and generator probes

Truncation is not a sufficiency test. Projecting a trained full-rank angle map onto its top singular directions after training costs 0.576 at rank 2 ($T = 128$), and rank 16 is needed to recover (0.992); yet a model trained at $r = 4$ reaches 0.919 at $T = 1024$. A map trained at a rank and one truncated to it are different objects.

Selective RoPE’s generator, knob by knob. Table 38. Every single-knob arm also deletes MapFormer’s frequency readout ($\text{diag}(\omega)W_{\text{out}}$ is replaced by a scalar), so the per-knob rows cannot be attributed to the knob; the rank sweep is unaffected by this confound. The two arms of about 8,000 extra parameters are indistinguishable from each other (+0.018, MDE 0.026, at $T = 512$; +0.028, MDE 0.045, at $T = 1024$).

The gate is not a token suppressor. In the gate arm, Selective RoPE’s gate passes actions more than observations by 1.35 $\times$ on the torus, where that arm gains, and passes one parity bit more than the other by 1.54 $\times$ on parity, where that arm loses. The pre-registered suppression hypothesis failed.

PoPE on top of path integration, by grid. MapPoPE minus MapWM, eight seeds per grid, one batch: at $T = 1024$ raw +0.139 (MDE 0.455), +0.122 (MDE 0.137), +0.101 (MDE 0.065, 8/8) and +0.123 (MDE 0.139) at grids 16, 32, 64 and 128. Grid 16 is bimodal (Vanilla converges on 2/8 seeds), so it is not used for the octave test. At grid 64, MapPoPE at $r = 4$ against $r = 2$ is +0.019 raw at $T = 1024$ (5/8; loss-matched +0.015), unmeasured.

Table 38: Selective RoPE components added to MapFormer-WM, minus **Vanilla**: parity at $L = 128$ ($n = 16$, trained at $L = 16$) and torus ($n = 8$, trained at $T = 128$), paired by seed. Extra parameters are on parity. Cells: delta (MDE, seeds +) or (paired t , seeds +); MDE = minimum detectable effect, $2.8\text{sd}/\sqrt{n}$; unmeasured = within it. Confounded attribution (see text): each single-knob arm also replaces MapFormer’s frequency readout. Reading: the full generator is not better on parity and marginally better on the torus at $T = 512$.

arm	extra params	parity	torus $T = 512$	torus $T = 1024$
causal convolution	+193	−0.020 (MDE 0.013)	−0.031 (MDE 0.048)	−0.064 (t −1.99)
no bottleneck	+7,873	−0.020 (MDE 0.018)	+0.022 (MDE 0.025)	+0.058 (t 2.13, 7/8)
sigmoid gate	+8,193	−0.030 (MDE 0.014)	+0.040 (MDE 0.030, 7/8)	+0.086 (t 3.05, 8/8)
all three	+16,385	−0.009, unmeasured	+0.031 (MDE 0.030, 7/8)	+0.048 (t 1.36)

Drift from a skewed basis. Integrating the learned linear map exactly along test trajectories, the residual after net displacement grows linearly in t for both $r = 2$ and $r = 4$, and is 11.38–11.51 times larger at $r = 2$ at every length. The pre-registered prediction from the opposition scores was 5.4; the link to accuracy is unmeasured (within-arm $r = -0.363$ at $n = 8$).

The forget gate over training. Peak decay occurs at 1% of training in six of eight seeds, and $r(\text{peak decay, gain}) = -0.531$, against a registered positive sign.

G EM kernel: supplementary

Kernel coherence set by construction. With q_0 and k_0 frozen at a fixed relative phase and matched magnitude (eight seeds per arm, one batch per task): on the torus at $T = 1024$, $|\rho| = 1$ minus $\rho = 0$ is +0.292 (MDE 0.063, 8/8), and $\rho = +1$ minus $\rho = -1$ is +0.016 (MDE 0.034), which has expectation zero because the content score absorbs the kernel’s sign. The coherence effect is established only for a frozen kernel at a peak amplitude of about 0.003, far below learned amplitudes (0.06–0.08 on the torus). On recency, $\rho = +1$ minus $\rho = 0$ is +0.191 (MDE 0.158, 7/8); the registered interaction between tasks ($\rho = +1$ minus $\rho = -1$, torus minus recency) is unmeasured (+0.058, MDE 0.182) and had expectation zero by the sign gauge; restated after the data on $|\rho|$ it is +0.113 (MDE 0.195), also unmeasured.

Phase degrees of freedom on the torus. At $n = 8$ at $T = 1024$: single origin 0.962 ± 0.024 , magnitude-only freedom 0.963 ± 0.022 , phase freedom 0.875 ± 0.112 , separate origins 0.809 ± 0.128 . Phase freedom minus magnitude-only is -0.088 (MDE 0.118), unmeasured; separate minus single is -0.154 (MDE 0.130, 0/8).

Recency, $n = 24$. Phase freedom (**EMDoF_alignfree**) minus a phase-locked control with magnitude free (**EMDoF_alignlock**) is +0.165 (MDE 0.088, 21/24) and +0.173 on fresh seeds (MDE 0.127); that control differs in optimiser dynamics, which the matched control of Section 8.7 removes.

The start and end of search. At initialisation the position pathway’s gradient is 1.7×10^{-4} to 4.3×10^{-4} of the content branch’s (8/8 seeds), the root-mean-square position score is $6\text{--}9 \times 10^{-4}$ against a content score of 0.31–0.34, and the cosine between the gradient and the direction that raises the kernel at the answer is -0.042 to $+0.023$. After training, the straight path from a token’s increment to its exact rewind has a median of 15.5 kernel peaks at $k = 60\text{--}64$ in single-origin MapEM, against 2 at initialisation; and for every $k \geq 32$ the rewind still scores higher on the kernel than where the token settled. These are two snapshots; nothing times when the barriers appear.

The 300-epoch offset-set batch. With k from a set of m at 300 epochs: $m = 4$ 1.000 ± 0.000 , $m = 16$ 0.996 ± 0.010 , $m = 64$ 0.578 ± 0.123 (primary readout); four times the budget for $m = 64$ gives 0.928 ± 0.131 (two seeds at 0.724 and 0.707). Of its two exposure-matched pairs, one was ceiling against ceiling and uninformative; the other was unmeasured (-0.068 , MDE 0.132, 1/8).

Kernel value at zero displacement. A pre-registered probe tested whether the sign or peakedness of MapEM’s kernel at zero displacement explains when MapEM wins; kernel quality was inversely related to accuracy across configurations, and the account was falsified. The sign gauge explains why a kernel negative at zero can still retrieve.

H State correction: supplementary

Refining the phase on the depth axis. A looped model that corrects θ on each pass ($\theta = \theta_0 + g \tanh(\text{refine}(x))$, gate initialised at zero with non-zero gradient) against the same loop with θ fixed, under action noise $p \in \{0, 0.10, 0.25\}$ ($n = 3$, exploratory): refine minus fixed is -0.001 , -0.011 and $+0.005$ at $T = 128$ and $+0.006$, $+0.003$ and -0.005 at $T = 512$, with no slope in noise against a registered positive slope.

Landmarks. On a torus with 200 single-use landmark cells (retrained under current code, $n = 3$, $T = 128$), context destruction passes (Level15 0.9850 to 0.1553 with actions resampled). A generic extra attention head with more parameters than the filter scores 0.9952 against Level15’s 0.9850 ($+0.010$, t 0.79), and Vanilla 0.8427, one of its seeds failing to converge. The landmark gain is therefore not evidence for the filter’s measurement mechanism ($n = 3$, exploratory).

Paper task and Match-Query at $n = 3$. On the paper task at 50 epochs Level15 scores 1.000 ± 0.000 against Vanilla 0.993 ± 0.009 (final training loss 0.0068 against 0.1126), a ceiling. On Match-Query, where the query phase gives the filter nothing to measure, it scores 0.8764 ± 0.2128 against 0.8884 ± 0.1401 . On the compositional task it improves likelihood on 3/3 seeds at both lengths while its transfer accuracy mean rests on one outlier seed.

I Loops: supplementary

Per-token recursion depth. On Match-Query, evaluating eight looped checkpoints at k passes gives 0.384, 0.835, 0.866, 0.865, 0.865, 0.863 and 0.856 for $k = 1, 2, 3, 4, 5, 6, 8$. An oracle that picks the best count for each fifth of the query phase scores 0.854 against 0.847 for the best single count ($k = 3$), both unweighted means over the five fifths, whereas the per-count values above pool tokens: at most $+0.007$, against a seed sd of 0.152. There is nothing for a per-position router to exploit.

Recursion buys an index model horizon (exploratory). On the paper task binned by recurrence interval ($n = 3$, 300 epochs), a four-pass loop on RoPE adds $+0.363$ (sd 0.018, MDE 0.029, 3/3) at interval 17–32, indistinguishable from four real layers (-0.023), at a quarter of their parameters. Every index configuration still falls to the floor beyond interval 32, where one path-integrated layer holds 0.945 at 65 or more.

Sampling the loop count. A loop trained at a fixed count of 4 peaks at 4 passes at $T = 128$ and at 2 at $T = 512$. Sampling the count per batch from 2–6 flattens the count curve: at one pass the sampled model scores 0.998 ± 0.003 at $T = 128$ where the fixed-count model scores 0.821 ± 0.191 . Its gain beyond training length is directional at $n = 5$, and it does not carry over to action noise.

J Language and borrowed benchmarks

MapFormer v4 on language. A 12-layer MapWM ($r = 4$) on OpenWebText at about 10^{11} tokens gives 18.79 ± 0.15 perplexity against RoPE’s 19.14 ± 0.14 over five seeds, no BLiMP gain (0.79 against 0.78), and perplexity that improves on RoPE at all tested lengths but degrades more sharply with length than the numbers reported for CoPE and PaTH.

enwik8. Byte-level, about 28.6M parameters, 36k iterations, sequence 512. MapPoPE minus RoPE is -0.0058 bits per byte (t 3.49, $n = 3$ each; final checkpoints). The single-axis arms are single-seed: PoPE-Flat 1.3746 and MapWM 1.3758 against RoPE 1.3799 and MapPoPE 1.3740. The position effect alone (-0.0041) is the size of the $n = 3$ MDE (0.0041), while the paper’s perplexity gain corresponds to about 0.0067 bits per byte, so this setup is underpowered, not null. A parameter-identical hierarchical/flat pair (28,371,016 parameters, $n = 1$) gives 1.4537 against 1.4506 bits per byte, $+0.0032$ for hierarchy, inside checkpoint noise of 0.003–0.007, at 20.10 against 16.36 iterations per second and 14.1% less peak memory.

Flip-Flop. Scoring only the final read (8 seeds, 4 layers), all arms but **CARoPE_r4** (0.03%) are at 0.00% error in distribution; on the dense out-of-distribution split path-integrated minus index is -1.23 percentage points (MDE 3.99) and monotone minus signed $+2.40$ (MDE 3.97), all unmeasured. The sparse split is solvable by reading the last bit (0.978).

MQAR. In four single-seed pilots at 2 layers, no arm formed an induction circuit: with four key-value pairs **Vanilla_r4** 0.273, **RoPE** 0.264 and **PlainFlat** 0.274 against a guessing rate of 0.250. The pre-registration predicted a ceiling. This is a feasibility floor, not a positional result.

K The MONOTONE batch

Pre-registered (**MONOTONE_PREREG.md**), 96 runs in one batch, 12 seeds per arm. Manipulation checks passed: every constrained checkpoint’s increments are non-negative on all seeds, every signed checkpoint’s go negative; at initialisation the signed MapEM twin is bitwise **VanillaEM_P0_r4**; and **Signed_r4** and **Abs_r4** seed 0 retrained here are bitwise identical to the stored sign batch (a determinism check, which is what licenses the eight-arm pool below).

Table 39: Experiment 1 (MapFormer-EM, recency) and Experiment 2 (Selective RoPE’s generator) arms of the pre-registered MONOTONE batch, $n = 12$ per arm, one batch: accuracy mean $\pm$ sd and final training loss; “–” = not evaluated. Recency = return the k -th most recent symbol, trained at $T = 1024$ (chance 0.0625); torus = paper task, accuracy predicting the observation at revisited cells of a held-out map, trained at $T = 128$ (floor 0.506). $\neg$ Abs = increments forced non-negative ($|\Delta|$), otherwise signed. **VanillaEM_P0_r4** = single-origin MapFormer-EM; **Signed_r4** = MapFormer-WM; **SRoPEGen** = Selective RoPE’s generator (gate, causal convolution, no bottleneck). Reading: on recency the constraint costs single-origin MapEM a detectable amount and the WM-type generators little; on the torus it costs **SRoPEGen** a large amount (Table 40).

task	arm	$T = 128$	$T = 512$	$T = 1024$	$T = 2048$	final loss
recency	VanillaEM_P0_r4	–	–	0.641 ± 0.136	0.543 ± 0.114	1.262
recency	EM_P0_Abs_r4	–	–	0.443 ± 0.117	0.375 ± 0.089	1.753
recency	Signed_r4	–	–	0.994 ± 0.020	0.898 ± 0.058	0.032
recency	Abs_r4	–	–	0.934 ± 0.098	0.883 ± 0.108	0.156
recency	SRoPEGen	–	–	0.944 ± 0.064	0.859 ± 0.071	0.181
recency	SRoPEGen_Abs	–	–	0.877 ± 0.068	0.806 ± 0.050	0.409
torus	SRoPEGen	1.000 ± 0.000	0.984 ± 0.008	0.899 ± 0.051	–	0.0003
torus	SRoPEGen_Abs	0.925 ± 0.028	0.638 ± 0.029	0.545 ± 0.029	–	0.3148

Registered verdicts. Experiment 1: **unresolved** (neither “subtraction needed”, which required P3 detectable, nor “wrap substitutes”, which required P1 not detectably negative, fired). Experiment 2: Q1 falsified as registered, Q2 falsified (the recency constraint costs a small detectable amount), Q3 met. Q1’s registered form could not detect the effect it tested, because its two-arm pool had non-overlapping final losses; a loss-matched contrast needs a pool in which loss varies independently of the manipulation. The eight-arm pool is post hoc; its $r(\text{loss}, \text{accuracy})$ is -0.742 at $T = 1024$ and -0.853 at $T = 512$. Rule 9 over the 72 recency runs is -0.987 .

Route (P4). Among solved cells ($\text{accuracy} \geq 0.9$, $k \geq 8$), the fraction retrieved through the query token’s own step, read as $\max(\text{sel} - \text{sel}_0, \text{selmin} - \text{selmin}_0) \geq 0.5$ in the best head: signed MapEM 0.953 (12/12 seeds

Table 40: MONOTONE contrasts: monotone (non-negative increments) minus signed accuracy, arms as in Table 39, paired by seed, $n = 12$, unless marked unpaired. P1–P3, Q1–Q3 = registered contrasts; rows under *exploratory* were not registered; “stored” = from the earlier sign ablation. sd = of paired differences (se = standard error); MDE = minimum detectable effect, $2.8 \text{sd}/\sqrt{n}$; detectable = beyond the MDE; unmeasured = within it, not evidence of zero; fractions = seeds with a positive difference. Loss-matched = compared at equal final training loss by pooled regression; Q1’s two-arm pool had non-overlapping losses, and the eight-arm pool adds the six stored sign arms. Reading: MapEM’s cost on recency is detectable, the registered interaction with MapWM’s cost is unmeasured, and raw on the torus SRoPEGen pays about MapWM’s cost.

contrast	delta	sd	MDE	verdict
P1 MapEM recency, monotone – signed, $T = 1024$	−0.198	0.192	0.155	detectable (2/12)
MapEM recency, $T = 2048$	−0.169	0.120	0.097	detectable (1/12)
P2 MapWM recency, monotone – signed, $T = 1024$	−0.060	0.104	0.084	unmeasured (1/12)
P3 interaction P1 – P2	−0.138	se 0.065	0.181	unmeasured
P1 loss-matched (reported, not read)	−0.043	0.063	0.051	unmeasured
torus SRoPEGen, monotone – signed, $T = 1024$, raw	−0.355	0.057	0.046	detectable (0/12)
Q1 the same, loss-matched, two-arm pool	−0.056	0.102	0.083	unmeasured; falsified as registered
Q2 recency SRoPEGen, monotone – signed, $T = 1024$	−0.068	0.067	0.054	detectable (2/12)
Q3 crossover, torus (raw) – recency	−0.287	se 0.025	0.071	detectable; met
<i>exploratory</i>				
torus SRoPEGen, loss-matched, eight-arm pool, $T = 1024$	−0.188	0.072	0.059	detectable (0/12)
torus MapWM, loss-matched, same pool, $T = 1024$	−0.273	0.083	0.067	detectable (0/12)
torus SRoPEGen, loss-matched, eight-arm pool, $T = 512$	−0.172	0.047	0.038	detectable (0/12)
torus MapWM raw, $T = 1024$ (stored)	−0.363	0.030	0.025	detectable (0/12)
SRoPEGen cost – MapWM cost, torus raw (unpaired)	+0.009	se 0.019	0.052	unmeasured
MapWM recency, $T = 2048$	−0.015	0.120	0.097	unmeasured (7/12)
SRoPEGen recency, $T = 2048$	−0.052	0.094	0.076	unmeasured (3/12)
MapEM solved cells ($k \geq 8$) per seed	−13.5	14.2	11.5	detectable (3/12)

with solved cells, 23.9 solved cells per seed), monotone MapEM 0.936 (11/12, 10.4). The monotone fraction averages over very few solved cells on some seeds: one cell on seed 7, three on seed 8, none on seed 10.

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